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Multi-joint link hinge and refrigerator having the same

Abstract

A multi-joint link hinge may include hinge bracket provided with a base in which a plurality of screw coupling holes are defined, a plurality of side walls which protrude from the base and are spaced apart from each other in a vertical direction, a plurality of link modules which are vertically arranged between the side portions and axially coupled to the hinge bracket by a rotation shaft, and a door bracket which is axially coupled to each of the plurality of link modules.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation application of U.S. patent application Ser. No. 17/320,591 filed May 14, 2021, which claims priority under 35 U.S.C. § 119 to Korean Application No. 10-2020-0059986 filed on May 19, 2020, whose entire disclosure is hereby incorporated by reference.

BACKGROUND

1 Field

(1) The present disclosure relates to a multi-joint link hinge used in furniture, home appliances, and a refrigerator.

2. Background

(2) Refrigerators are home appliances that store food or other items (hereafter “food”) at a low temperature. A storage space inside the refrigerator where the food is placed may be covered by a door and be cooled using cool air. The cool air may be generated by being heat-exchanged with a refrigerant circulated in a refrigeration cycle.

(3) The storage space may be partitioned into an upper storage space at an upper side and a lower storage space at a lower side, which may be opened and closed by an upper door and a lower door, respectively. An upper end and a lower end of each of the upper and lower doors may be supported by a hinge device to be rotatably mounted such that the upper and lower storage spaces may be opened and closed by rotation of the upper and lower doors.

(4) Korean Patent Publication No. 10-2016-0099982 discloses an intermediate hinge provided on a front surface of a main body, and a rotation shaft that vertically extends from a horizontal support portion that protrudes between an upper door and a lower door to support a lower end of the upper door and an upper end of the lower door. Since an interference between adjacent objects (e.g., a wall surface or furniture) may occur when the door rotates due to a thickness of the door, the refrigerator may be installed to protrude more than the adjacent lateral objects or to be spaced a predetermined distance from the adjacent lateral objects to reduce or prevent interference. The refrigerator having such a structure may have limited installation configurations that do not provide a sense of unity with the adjacent surroundings. For example, the refrigerator having the above structure may not be able to be configured as a built-in type or otherwise installed to be continuous with adjacent furniture or home appliances.

(5) To solve this limitation, a hinge device has been developed. The developed hinge device rotates so as not to interfere with adjacent objects when the door of the refrigerator rotates. Korean Patent Registration No. 10-1497295 discloses a multi-link hinge having a rotation structure that has multi-joints such that a door mounted with the multi-link hinge does not interfere with adjacent objects. However, the disclosed structure is relatively thick and has a large width, and when a pair of hinges to support upper and lower doors are provided between upper and lower doors of a refrigerator, due to a size of the hinge, a capacity of each of the upper and lower storage spaces may be reduced. A vertical interval between the upper and lower doors may be widened, and the hinge may be exposed due to this wide interval, which comprises an appearance.

(6) The above references are incorporated by reference herein where appropriate for appropriate teachings of additional or alternative details, features and/or technical background.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The embodiments will be described in detail with reference to the following drawings in which like reference numerals refer to like elements wherein:

(2) FIG. 1 is a perspective view illustrating a state in which a refrigerator is installed according to an embodiment.

(3) FIG. 2 is a perspective view illustrating a state in which an upper door and a lower door of the refrigerator are opened.

(4) FIG. 3 is a partial perspective view illustrating a state in which an upper hinge is mounted according to an embodiment.

(5) FIG. 4 is a view illustrating a state of the door, a wall, and the upper hinge in a state in which the upper door is closed.

(6) FIG. 5 is a view illustrating a state of the door, the wall, and the upper hinge in a state in which

the upper door rotates at a set angle.

(7) FIG. 6 is a view illustrating a state of the door, the wall, and the upper hinge in a state in which the upper door is opened more than in FIG. 5.

(8) FIG. 7 is a view illustrating a state of the door, the wall, and the upper hinge in a state in which the upper door is opened more than in FIG. 6.

(9) FIG. 8 is a view illustrating a state of the door, the wall, and the upper hinge in a state in which the upper door is fully opened.

(10) FIG. 9 is a partial perspective view illustrating a state in which a lower hinge is mounted according to an embodiment.

(11) FIG. 10 is a partial perspective view illustrating a state in which a center hinge is mounted according to an embodiment.

(12) FIG. 11 is a plan view illustrating a state in which the center hinge is folded.

(13) FIG. 12 is a perspective view illustrating the state in which the center hinge is folded when viewed from an upper side.

(14) FIG. 13 is a perspective view illustrating the state in which the center hinge is folded when viewed from a lower side.

(15) FIG. 14 is a perspective view illustrating a state in which the center hinge is unfolded at a predetermined angle.

(16) FIG. 15 is a front view of the center hinge.

(17) FIG. 16 is an exploded perspective view illustrating a state in which the center hinge is disassembled when viewed from one side.

(18) FIG. 17 is an exploded perspective view illustrating a state in which the center hinge is disassembled when viewed from the other side.

(19) FIG. 18 is a cross-sectional view taken along line XVIII-XVIII' of FIG. 11.

(20) FIG. 19 is a cross-sectional view taken along line XIX-XIX' of FIG. 11.

(21) FIG. 20 is an exploded perspective view of an upper link module that is one component of the center hinge when viewed from the upper side.

(22) FIG. 21 is an exploded perspective view of the upper link module when viewed from the lower side.

(23) FIG. 22 is an exploded perspective view of a lower link module that is one component of the center hinge when viewed from the upper side.

(24) FIG. 23 is an exploded perspective view of the lower link module when viewed from the lower side.

(25) FIG. 24 is a view illustrating a state in which the center hinge is mounded when viewed from one side.

(26) FIG. 25 is a cross-sectional view taken along line XXX-XXX' of FIG. 1.

(27) FIG. 26 is a cross-sectional view taken along line XXVI-XXVI' of FIG. 24.

(28) FIG. 27 is a view illustrating a state of the center hinge in a state in which all the upper door and the lower door are closed.

(29) FIG. 28 is a view illustrating a state of the center hinge in a state in which the upper door is opened at a set angle.

(30) FIG. 29 is a view illustrating a state of the center hinge in a state in which the upper door is opened more than in FIG. 28.

(31) FIG. 30 is a view illustrating a state of the center hinge in a state in which the upper door is opened more than in FIG. 29.

(32) FIG. 31 is a view illustrating a state of the center hinge in a state in which the upper door is fully opened.

(33) FIG. 32 is a view illustrating a state of the center hinge in a state in which the lower door is opened at a set angle.

(34) FIG. 33 is a view illustrating a state of the center hinge in a state in which the lower door is

opened more than in FIG. 32.

(35) FIG. 34 is a view illustrating a state of the center hinge in a state in which the lower door is opened more than in FIG. 33.

(36) FIG. 35 is a view illustrating a state of the center hinge in a state in which the lower door is fully opened.

(37) FIG. 36 is a perspective view of a center hinge according to another embodiment.

(38) FIG. 37 is a perspective view of a center hinge according to further another embodiment.

(39) FIG. 38 is a perspective view of a refrigerator according to another embodiment.

(40) FIG. 39 is a perspective view illustrating a state in which an upper door and a lower door of the refrigerator are opened.

DETAILED DESCRIPTION

(41) Referring to FIGS. 1-2, a refrigerator 1 according to an embodiment may include a cabinet 10 defining a storage space. The storage space may have an opened front surface, and a door may open or close the storage space at the opened front surface. An outer appearance of the refrigerator 1 may be defined by the cabinet 10 and/or doors 20 and 30.

(42) The refrigerator 1 may be installed or mounted so as to blend in with furniture and/or a wall of an indoor space. For example, as illustrated in FIG. 1, the refrigerator 1 may be installed in an indoor space such as a kitchen and may be provided adjacent to built-in kitchen cabinets, closets, and/or the wall to harmonize an appearance. A front surface of the refrigerator 1 may be installed to be flush with front surfaces of the built-in cabinets and/or wall to create a continuous or uniform appearance.

(43) For convenience of description, the surrounding furniture, built-in cabinets, or wall will be referred to as surrounding or adjacent furniture O. One of ordinary skill in the art will understand that the surrounding furniture O could alternatively be a wall or refer to kitchen cabinets, pantry doors, smaller walls or fences, counters, or other home appliances, fixtures, or furniture.

Embodiments disclosed herein are not limited by a configuration of the surrounding furniture O.

(44) A space or recess corresponding to a size of the refrigerator 1 may be provided in the surrounding furniture O, and the refrigerator 1 may be provided in the space to be a built-in type refrigerator. As another alternative, a plurality of refrigerators 1 may be continuously provided, or other home appliances may be continuously provided, in addition to or as a replacement to the surrounding furniture O.

(45) Front surfaces of the doors 20 and 30 may be very close to front surfaces or sides of the surrounding furniture O and be provided on a same or adjacent plane to create a uniform appearance. The front surfaces of the doors 20 and 30 may be made of the same or similar material (e.g., a material having the same texture) as the surrounding furniture O to further enhance a sense of unity.

(46) The cabinet 10 may define a storage space that is partitioned vertically to create an upper storage space 12 and a lower storage space 13, but embodiments disclosed herein are not limited to refrigerators that are vertically split. A barrier 11 (e.g., a partition wall or shelf) may define the upper storage space 12 above the barrier 11 and the lower storage space 13 below the barrier 11. For example, the upper storage space 12 may be used as a refrigerating compartment, and the lower storage space 13 may be used as a freezing compartment and maintained at a temperature below the a temperature of the refrigerating compartment. The upper storage space 12 may alternatively be referred to as a refrigerating compartment, and the lower storage space 13 may alternatively be referred to as a freezing compartment.

(47) Embodiments disclosed herein are not limited to cabinets 10 divided into upper and lower storage spaces 12 and 12. For example, the cabinet 10 may be horizontally partitioned by a vertical barrier or partition wall into left and right storage spaces. As another alternative, the cabinet 10 may be split into three or more spaces. Embodiments disclosed herein are not limited to a partitioning configuration of the cabinet.

(48) The doors **20** and **30** may include at least one upper door **20** and at least one lower door **30**. The upper door **20** may be rotatably mounted on the cabinet **10** to open and/or close the upper storage space **12**. The lower door **30** may be rotatably mounted on the cabinet **10** to open and close the lower storage space **13** by the rotation thereof. When the cabinet **10** is partitioned to have left and right storage spaces, the doors **20** and **30** may alternatively include left and right doors **20** and **30** to open and/or close the left and right storage spaces, respectively.

(49) Upper and lower ends of the upper door **20** may be supported by an upper hinge **40** and a multi-joint link hinge **60**, respectively. The upper door **20** may rotate by the upper hinge **40** and the multi-joint link hinge **60** to open and/or close the upper storage space **12**.

(50) There may be two upper doors **20**. For example, the two upper doors **20** may include a left upper door **20a** and a right upper door **20b** (where “left” and “right” have the orientation as shown in FIG. **4**) which each independently rotate to open and/or close left and right sides, respectively, of the upper storage space **12**. There may be two upper hinges **40** and two multi-joint link hinges **60** corresponding to the two upper doors **20**, respectively. The two upper hinges **40** may include a left upper hinge provided at a left side and rotatably mounted to the cabinet **10** and a right upper hinge provided at a right side and rotatably mounted to the cabinet **10**. The two multi-joint link hinges **60** may include a left multi-joint link hinge provided at a left side and rotatably mounted to the cabinet **10** and a right multi-joint link hinge provided at a right side and rotatably mounted to the cabinet **10**.

(51) A dispenser **201** configured to dispense water or ice may be provided at an outer side of the upper door **20** so as to dispense water or ice when the upper door **20** is closed. The dispenser **201** may be installed or located on a front surface of the upper door **20** and configured to receive liquid (e.g., water) from a liquid supply.

(52) Upper and lower ends of the lower door **30** may be supported by the multi-joint link hinge **60** and a lower hinge **50**, respectively. The lower door **30** may rotate by the multi-joint link hinge **60** and the lower hinge **50** to open and/or close the lower storage space **13**.

(53) There may be two lower doors **30**. For example, the two lower doors **30** may include a left lower door **30a** and a right lower door **30b** (where “left” and “right” correspond to the left and right upper doors **20a** and **20b** in FIG. **4**) which each independently rotate to open and/or close left and right sides, respectively, of the lower storage space **13**. There may be two lower hinges **50** and two multi-joint link hinges **60** corresponding to the two lower doors **30**, respectively. The two lower hinges **50** may include a left lower hinge provided at a left side and rotatably mounted to the cabinet **10** and a right lower hinge provided at a right side and rotatably mounted to the cabinet **10**. The two multi-joint link hinges **60** may include a left multi-joint link hinge provided at a left side and rotatably mounted to the cabinet **10** and a right multi-joint link hinge provided at a right side and rotatably mounted to the cabinet **10**.

(54) A handle or opening/closing space **301** may be between a lower end of the upper door **20** and an upper end of the lower door **30**. A separate handle (e.g., a recess or protrusion) to open and/or close the upper door **20** and/or the lower door **30** may be provided to be accessible from the handle space **301**, e.g., at a bottom surface of the upper door **20** and/or a top surface of the lower door **30**. Such a handle may be recessed in a groove shape. As an alternative, handles (e.g., bars) may be formed on each of the upper and lower doors **20** and **30**.

(55) The upper hinge **40**, the multi-joint link hinge **60**, and the lower hinge **50** may rotate in a same or similar trajectory. The upper door **20** and the lower door **30** may smoothly rotate without interfering with the surrounding furniture **O** during opening and closing.

(56) Referring to FIGS. **3-8**, the upper hinge **40** may be mounted at a corner defined by an upper front end and a side end of the cabinet **10**. The upper hinge may be connected to an end of a top surface of the upper door **20**.

(57) The upper hinge **40** may have a structure in which a plurality of links are coupled to each other, and thus, when the upper hinge **40** rotates, the upper door **20** may rotate while moving in a

direction away from the front surface of the cabinet **10**.

(58) The rotation trajectory of the upper door **20** may be determined by the structure of the plurality of links constituting the upper hinge **40**. The structure of the upper hinge **40** may be configured such when left and right upper doors **20a** and **20b** are provided side by side, and the surrounding furniture **O** is provided at sides of the left and right upper doors **20a** and **20b**, an opening and/or closing of the left and right upper doors **20a** and **20b** may not interfere with the surrounding furniture **O**. The upper hinge **40** may be referred to as a multi-link hinge.

(59) The upper hinge **40** and the lower hinge **50** may have a same or similar structure. The multi-joint link hinge **60** may have a structure different from that of each of the upper and lower hinges **40** and **50**, but may be configured to have a same rotational trajectory as the upper and lower hinges **40** and **50**. Such rotation trajectories of the upper, lower, and multi-joint link hinges **40**, **50**, and **60** may be configured so that rotation of the upper door **20** and the lower door **30** may be smoothly performed along a same trajectory.

(60) The upper hinge **40** may include a hinge bracket **41** mounted on the cabinet **10**, a main link **42** axially coupled to the hinge bracket **41**, a first sub link **43** and a second sub link **44**, which are axially coupled to the main link **42**, and a door bracket **45** which is axially coupled to ends of the first sub link **43** and the second sub link **44** and is coupled to a top surface of the upper door **20**.

(61) Each of the links **42**, **43**, **44**, and **45** may be axially coupled to define a quadrilateral shape as a whole and may be folded or unfolded to provide a trajectory through which the upper door **20** rotates. The hinge bracket **41** and the second sub link **44** may be connected to each other by a linear damper **46** (FIG. 5) having ends that are axially coupled to each other. The linear damper **46** may reduce rotation when the upper hinge **40** is folded to close the upper door **20** to alleviate an impact.

(62) The first sub link **43** may be provided with a spring **47** (FIG. 6) that is tensioned or compressed according to a rotation of the first sub link **43** to force the rotation of the first sub link **43**. The spring **47** may be, for example, a compression spring or a tension spring. The spring **47** may be compressed while the upper door **20** is closed and may be restored immediately before the upper door **20** is completely or fully closed. The spring **47** may assist a rotation of the first sub link **43** at the moment the upper door **20** is closed by the spring **47**. The upper door **20** may be effectively closed even when the linear damper **46** operates. The plurality of links **42**, **43**, **44**, and **45** of the upper hinge **40** may rotate while maintaining a set trajectory by an action of the linear damper **46** and the spring **47**.

(63) Referring to FIG. 4, the pair of upper doors **20** may be provided side by side at a front of the upper storage space **12**. The front surface of the upper door **20** may be spaced a prescribed or predetermined interval or distance **W1** from the surrounding furniture **O**, which may be provided at both sides of the refrigerator **1**. For example, the predetermined distance **W1** may be about 3 mm. While ensuring an initial rotation of the upper door **20** so as not to interfere, when the upper door **20** is closed, a space between the upper door **20** and the surrounding furniture **O** may be narrowed to enhance the sense of unity.

(64) The upper storage space **12** may be shielded by the pair of upper doors **20**, and the left upper door **20a** and the right upper door **20b** may independently rotate by the upper hinges **40**, respectively. The left upper door **20a** and the right upper door **20b** may have a rotational structure independent from each other. A distance or space between the left upper door **20a** and the right upper door **20b** may be a predetermined or prescribed interval **W2** so that the left upper door **20a** does not interfere with a rotation of the right upper door **20b** during rotation and vice versa. For example, the predetermined interval **W2** may be about 7 mm to about 8 mm. The predetermined interval may be set differently according to a thickness of the upper door **20**.

(65) A pillar **203** (or alternatively, filter material) may be provided between the left upper door **20a** and the right upper door **20b** to shield the space between the pair of upper doors **20**. The pillar **203** may be rotatably mounted on, for example, the right upper door **20b** (or alternatively, the left upper door **20a**) and may be unfolded by a guide provided in the cabinet **10** while the right upper door

20b is closed. As illustrated in FIG. 4, when the right upper door **20b** is fully closed, the pillar **203** may be unfolded to shield the space between the left upper door **20a** and the right upper door **20b**. (66) While the right and/or left upper doors **20b** and/or **20a** is opened, the pillar **203** may be folded as illustrated in FIG. 5 by the guide provided in the cabinet **10**. While the right upper door **20b** rotates, the right upper door **20b** may not interfere with the left upper door **20a** or the surrounding furniture **O**.

(67) The upper hinge **40** may rotate from a fully folded state, shown in FIG. 4, to a partially unfolded state, shown in FIG. 5, according to the rotation operation of the upper door **20** (e.g., the right upper door **20b**). An operation of the right upper door **20b** may be described or referred to for convenience of description, but the left upper door **20a** and the upper hinge **40** connected to the left upper door **20a** may have a same or similar structure to have a same or similar operation.

(68) When the upper door **20** is opened at a first predetermined or prescribed angle between the front surface of the upper door **20** and the front surface of the cabinet **10** (e.g., 13°), the upper door **20** may rotate while moving forward. The pillar **203** of the right upper door **20b** may move to a front side of the left upper door **20a** without interfering with an end or side of the left upper door **20a**. A left end of the right upper door **20b** may also be maintained to be spaced apart from the surrounding furniture **O** so as not to interfere. The main link **42**, the first sub link **43**, the second sub link **44**, and the door bracket **45**, which constitute the upper hinge **40**, may start to rotate.

(69) As illustrated in FIGS. 6 and 7, the upper door **20** may gradually rotate to be opened. Here, the main link **42**, the first sub link **43**, the second sub link **44**, and the door bracket **45** may rotate so that the end of the upper door **20** rotates so as not to interfere with the surrounding furniture **O**. Even when the upper door **20** rotates to a second and/or third predetermined angle (e.g., about 30° in FIG. 6 and/or about 90° in FIG. 7), the upper door **20** may rotate by the upper hinge **40** so as not to interfere with the surrounding furniture **O**.

(70) As illustrated in FIG. 8, the upper door **20** may rotate up to a maximum open state at a fourth or “opened state” predetermined angle (for example, 130°). When the upper door **20** is fully opened, access to storage members (e.g., shelves and drawers) inside the cabinet **10** is easy. When the storage member (e.g., drawer) is pulled in and out, the storage member may be configured rotate up to an angle at which the storage member does not interfere with the doors **20** and **30**.

(71) When the upper door **20** fully rotate to be opened at the opened state predetermined angle, a structure such as a door dike protruding along a circumference of a rear surface of the upper door **20** may not interfere with an insertion or withdrawal of a drawer or other insertable storage member. When the upper door **20** fully rotates, the main link **42**, the first sub link **43**, the second sub link **44**, and the door bracket **45** may rotate so that the end of the upper door **20** rotates so as not to interfere with the surrounding furniture **O**. A distance between the end of the upper door **20** and a front surface of the surrounding furniture **O** may be spaced a predetermined or prescribed interval **W3** (e.g., 9 mm) from each other.

(72) As described above, when the upper door **20** rotates, the upper door **20** may rotate by the upper hinge **40** along a corner of the surrounding furniture **O** while maintaining the predetermined interval **W3** so as not to interfere with the corner of surrounding furniture **O**. If the upper door **20** and the surrounding furniture **O** are too far from each other, an interference may occur between the adjacent upper doors **20**, and a user's finger or body may be caught between the upper door **20** and the surrounding furniture **O**.

(73) The predetermined interval **W3** between the upper door **20** and the surrounding furniture **O** may be maintained to be about 3 mm to about 6 mm. The upper hinge **40** may be configured in a combination of the plurality of coupled link structures, the spring **47**, and the linear damper **46** so that the upper door **20** rotates along a predetermined or prescribed trajectory while being maintained at a predetermined or prescribed space from the adjacent upper door **20** and the surrounding furniture **O** when opening and closing the upper storage space **12**.

(74) Hereafter a lower hinge **50** rotatably supporting the lower door **30** will be described with

reference to the drawings. Referring to FIG. 9, the lower hinge **50** may be mounted on a front lower end of the cabinet **10** to rotatably support a lower end of the lower door **30** at a lower side.

(75) The lower door **30** may be provided in pair like the upper door **20** at both left and right sides. A width of the lower door **30** may be the same as or similar to that of the upper door **20**, and the front surface of the refrigerator **1** may appear symmetrical when viewed from a front side.

(76) The lower door **30** may rotate along a same or similar trajectory as the upper door **20**. The lower hinge **50** for the rotation of the lower door **30** may have substantially the same or similar structure as the upper hinge **40**. The lower hinge **50** may include a hinge bracket **51** mounted on the cabinet **10**, a main link **52** axially coupled to the hinge bracket **51**, a first sub link **53** and a second sub link **54**, which are axially coupled to the main link **52**, and a door bracket **55** which is axially coupled to ends of the first sub link **53** and the second sub link **54** and is coupled to a bottom surface of the lower door **30**.

(77) Each of the links **52**, **53**, and **54** may be axially coupled to define a quadrilateral shape as a whole and may be folded or unfolded to provide a trajectory through which the lower door **30** rotates. The hinge bracket **51** and the second sub link **54** may be connected to each other by a lower linear damper having both ends that are axially coupled to each other. The lower linear damper may reduce rotation when the lower hinge **50** is folded to close the lower door **30**, reducing an impact.

(78) The first sub link **53** may be provided with a spring that is tensioned or compressed according to a rotation of the first sub link **53** to force the rotation of the first sub link **53**. The spring may be, for example, a compression spring or a tension spring. The spring may be compressed while the lower door **30** is closed and may be restored immediately before the lower door **30** is completely closed. The spring may assist the rotation of the first sub link **53** at the moment at which the lower door **30** is closed by the spring **47**. The lower door **30** may be effectively closed even when the lower linear damper **46** operates. The plurality of links **52**, **53**, **54**, and door bracket **55** constituting the lower hinge **50** may rotate while maintaining a set trajectory by the action of the lower linear damper **46** and the spring.

(79) Referring to FIGS. **10-15**, the multi-joint link hinge **60** may be provided on the front surface of the cabinet **10**. Each of the bottom surface of the upper door **20** and the top surface of the lower door **30** may be independently rotatably supported by the multi-joint link hinge **60**.

(80) Since the multi-joint link hinge **60** may be positioned between the upper hinge **40** and the lower hinge **50**, the multi-joint link hinge **60** may be referred to as a center hinge **60**. The multi-joint link **60** may be applied to furniture or home appliances in which doors are vertically arranged, and is not limited to a structure of the refrigerator.

(81) The multi-joint link hinge **60** may rotatably support the upper door **20** and the lower door **30**. The multi-joint link hinge **60** may be provided in a space between the bottom surface of the upper door **20** and the top surface of the lower door **30** and may be mounted on the front surface of the cabinet **10**. The multi-joint link hinge **60** may be mounted on a front surface of the barrier **11** and be connected to the bottom surface of the upper door **20** and the top surface of the lower door **30**, respectively, so that the upper door **20** and the lower door **30** rotate independently.

(82) The multi-joint link hinge **60** may include a hinge bracket **61** (The hinge bracket **61** may alternatively be referred to as a first bracket) mounted on the barrier **11**, an upper link module or assembly **70** (The upper link module **70** may alternatively be referred to as a first linkage), and a lower link module or assembly **80** (The lower link module **70** may alternatively be referred to as a second linkage). The upper and lower link assemblies **70** and **80** may include a plurality of links that are axially coupled to the hinge bracket **61**. The upper link assembly **70** may be coupled to the bottom surface of the upper door **20**. The lower link assembly **80** may be coupled to the top surface of the lower door **30**. The upper and lower link assemblies **70** and **80** may alternatively be referred to as upper and lower linkages.

(83) The upper link assembly **70** and the lower link assembly **80** may be vertically provided in an inner region of the hinge bracket **61**. A thickness of the multi-joint link hinge **60** may be

determined by a vertical width of the hinge bracket **61**, and a sum of the vertical widths of the upper link assembly **70** and the lower link assembly **80** may be equal to or less than a sum of the vertical widths of the hinge bracket **61**.

(84) The upper link assembly **70** may have a structure that supports the upper door **20** at a lower side, and the upper door **20** may apply a load to upper link assembly **70**, which may cause drooping or deformation of the upper door **20**. The lower link assembly **80** may have a structure that supports the lower door **30** at an upper side, and the lower door **30** may apply a relatively low load. The lower door **30** may be supported by the lower hinge **50**.

(85) The upper link assembly **70** may have a thickness greater than that of the lower link assembly **80** to prevent the upper link assembly **70** from drooping or being deformed downward by the load of the upper door **20**. A limited width of the multi-joint link hinge **60**, i.e., the vertical width of the upper link assembly **70** within the vertical width of the hinge bracket **61** may be designed to be greater than the vertical width of the lower link assembly **80** to prevent the upper link assembly **70** from drooping or being deformed.

(86) The vertical width of the hinge bracket **61** may be determined according to the vertical width of the barrier **11**. The barrier **11** may be designed to have a thickness at which the upper storage space **12** and the lower storage space **13** are partitioned in an adiabatic state.

(87) A portion of an upper door gasket **21** and a lower door gasket **31**, which may be provided along rear circumferences of the upper door **20** and the lower door **30**, may not interfere with the multi-joint link hinge **60** and be in close contact with the front surface of the barrier **11**. A heating member **113** may be provided along a circumference of the upper storage space **12** and a circumference of the lower storage space **13** in addition to the barrier **11**. The heating member may be provided as a hot gas pipe or a heater.

(88) The heating member **113** may be configured to prevent condensation from being generated on the front surface of the cabinet **10** and may be provided inside the barrier **11** to extend to left and right sides. The heating member **113** may be provided along each of the upper storage space **12** and the lower storage space **13** and thus may be spaced apart from each of upper and lower portions of the hinge bracket **61**.

(89) The hinge bracket **61** may be mounted on the front surface of the barrier **11**. The hinge bracket **61** may be provided between the heating members **113** that are provided vertically and may be provided between the upper door gasket **21** and the lower door gasket **31**.

(90) As the vertical width of the hinge bracket **61** may be reduced or minimized, an interval between the heating members **113**, which may be arranged vertically, and an interval between the upper door gasket **21** and the lower door gasket **31** may be designed to be narrower. A thickness of the barrier **11** may be also reduced to increase storage capacity of each of the upper storage space **12** and the lower storage space **13**.

(91) The vertical width of the multi-joint link hinge **60** may be reduced or minimized to allow a range of movement for opening and closing of the upper door **20** and the lower door **30**. To minimize or reduce the vertical width of the multi-joint link hinge **60**, the upper link assembly **70** and the lower link assembly **80** may be provided together on one hinge bracket **61**. The multi-joint link hinge **60** may have a compact structure to occupy a relatively small or minimum space between the cabinet **10**, the upper door **20**, and the lower door **30**.

(92) When the upper door **20** and the lower door **30** are closed so that both the upper link assembly **70** and the lower link assembly **80** are fully folded, as illustrated in FIGS. **11** to **13**, the multi-joint link hinge **60** may have an approximately hexahedral shape and also may have a shape in which the plurality of links constituting the hinge bracket **61**, the upper link assembly **70**, and the lower link assembly **80** correspond to each other or fit together in a folded state so as not to interfere with each other.

(93) The upper link assembly **70** and the lower link assembly **80** may be configured to have the same or similar rotation trajectory, but may differ in thickness and coupling structure, and lengths

and coupling structures of the links may be the same as or similar to each other.

(94) To stably open and close the upper door **20** and the lower hinge **50**, the multi-joint link hinge **60** may have the same rotation trajectory as each of the upper hinge **40** and the lower hinge **50**. Configurations of the upper link assembly **70** and the lower link assembly **80** constituting the multi-joint link hinge **60** may correspond to the configurations of the upper hinge **40** and the lower hinge **50**, respectively.

(95) The upper link assembly **70** and the lower link assembly **80** may be mounted to the hinge bracket **61** to have the same rotation axis, but may rotate independently. The main rotation shaft **62** and the sub rotation shaft **63** mounted on the hinge bracket **61** may be provided to pass through both the upper link assembly **70** and the lower link assembly **80**, and the upper link assembly **70** and the lower link assembly **80** may be spaced apart from each other so that the upper link assembly **70** and the lower link assembly **80** do not interfere with each other during the rotation.

(96) The structure of each of the hinge bracket **61**, the upper link assembly **70**, and the lower link assembly **80**, which constitute the multi-joint link hinge **60**, in addition to unexplained reference numerals, will be described in more detail with reference to the drawings.

(97) Referring to FIGS. **16-17**, the hinge bracket **61** may be provided in a plate shape made of a metal material, and the top and bottom surfaces may be bent to define an accommodation or receiving space **610** in which the upper link assembly **70** and the lower link assembly **80** are accommodated.

(98) The hinge bracket **61** may include a base or plate **611** that is fixed to the cabinet **10** and a side or wall **612** that is bent to extend forward from each of upper and lower ends of the base **611**.

(99) The base **611** may be provided in a plate shape, and a plurality of coupling holes **613** may be provided in the base **611**. A coupling member such as a screw **615** (FIG. **10**) may be inserted into the plurality of coupling holes **613** to allow the hinge bracket **61** to be fixed and mounted on the front surface of the cabinet **10**.

(100) The side **612** may be bent perpendicularly from the base **611** to extend forward. As the side **612** extends from one end to the other end, a width in the extension direction may increase. A main side hole **616** and a sub side hole **617** may be provided in the side **612**. A main rotation shaft **62** and a sub rotation shaft **63** may pass through the main side hole **616** and sub side hole **617**, respectively.

(101) The side **612** may have a narrower width in an inner direction (left side in FIG. **16**) toward the center of the inside of the cabinet **10** with respect to a center of the hinge bracket **61**, and have a wider width in an outer direction (right side in FIG. **16**) of the cabinet **10** toward a side surface of the cabinet **10**. The main side hole **616** and the sub side hole **617** may be provided outside the side **612** of which a width is relatively wide to provide a structure in which the upper link assembly **70** and the lower link assembly **80** are rotatable. The sub side hole **617** may be provided at an outer end of the side **612** and may be provided further in front of the main side hole **616**.

(102) A shape of the extending front end of the side **612** may correspond to a shape of each of the upper link assembly **70** and the lower link assembly **80**, and thus, the upper link assembly **70** and the lower link assembly **80** may not interfere with each other even in the fully folded state.

(103) A base opening **618** may be provided in the base **611** and correspond to the position of the main rotation shaft **62**. An interference between ends of an upper main link **71** and a lower main link **81** penetrated by the main rotation shaft **62** may be avoided by the base opening **618**.

(104) The multi-joint link hinge **60** may be mounted to protrude more than the upper hinge **40** and the lower hinge **50** due to an influence of the heating member **113** provided inside the barrier **11**. The base opening **618** may be provided so that an axial coupling position of the upper link assembly **70** and the lower link assembly **80** are closer to the front surface of the barrier **11**, and an operation may not interfere with the hinge bracket **61**. Despite the protruding structure of the front surface of the barrier **11**, the upper link assembly **70** and the lower link assembly **80** of the multi-joint link hinge **60** may have the same or similar rotation trajectory as the upper hinge **40** and the lower hinge **50**.

(105) A shielding portion or cover **614** may be further provided on an outer end of the base **611**. The shielding portion **614** may be bent forward from the outer end of the base **611** and may be coupled to an outer end of the side **612**. A portion of an outer surface of the hinge bracket **61** may shield a side of the receiving space **610** to prevent the upper link assembly **70** and the lower link assembly **80** from being exposed to a side of the hinge bracket **61**.

(106) The shielding portion **614** may connect the base **611** to the side **612** so that strength of the hinge bracket **61** is further reinforced. An extending end of the shielding portion **614** may be provided behind the sub side hole **617**. The shielding portion **614** may prevent the upper link assembly **70** and the lower link assembly **80** from interfering with each other when the upper link assembly **70** and the lower link assembly **80** rotate.

(107) The upper link assembly **70** and the lower link assembly **80** may be vertically arranged and be penetrated by the main rotation shaft **62** and the sub rotation shaft **63** to rotate independently. The upper link assembly **70** and the lower link assembly **80** may be provided inside the receiving space **610** to be close to, but vertically spaced apart from, each other so as not to be caught and/or restricted during the rotation operation.

(108) The main rotation shaft **62** may pass through the main side hole **616** and sequentially pass through the upper main link **71** of the upper link assembly **70** and the lower main link **81** of the lower link assembly **80**. Each of the upper and lower main links **71** and **81** may be spaced apart from each other by a forming or contact portion **712a**. The forming portion **712a** may alternatively be referred to as a spacer or protrusion.

(109) The forming portion **712a** may be formed to protrude by forming a portion of a bottom surface of the upper main link **71** through which the main rotation shaft **62** passes. The forming portion **712a** may protrude to be in contact with a top surface of the lower main link **81**, and remaining portions of the bottom surface of the upper main link **71** and the top surface of the lower main link **81** except for the forming portion **712a** may be maintained to be spaced apart from each other.

(110) A main rotation shaft spacer **817** may be further provided below the lower main link **81**. The lower main link **81** may be spaced apart from the side **612** and may be rotatable without interfering with the side **612**.

(111) The sub rotation shaft **63** may pass through the sub side hole **617** and may sequentially pass through an upper connection link **72** of the upper link assembly **70** and a lower connection links **82** of the lower link assembly **80**. Each of the upper and lower connection links **72** and **82** may be spaced apart from each other by a sub rotation shaft spacer **64**.

(112) The sub rotation shaft spacer **64** may be provided between the side portions **612** and be penetrated by the sub rotation shaft **63**. A spacer groove **641** may be provided in a circumferential surface of the sub rotation shaft spacer **64** between an upper end and a lower end of the sub rotation shaft spacer **64**. An end of the upper connection link **72** may be inserted into the spacer groove **641**, and the upper connection link **72** may be penetrated by the sub rotation shaft **63** and inserted into the spacer groove **641**. A vertical height of the upper connection link **72** may be maintained inside the upper main link **71** by the sub rotation shaft spacer **64**, and while the upper link assembly **70** operates, the upper connection link **72** may not interfere with the upper main link **71** but operate inside the upper main link **71**.

(113) The lower connection link **82** may be provided below the sub rotation shaft spacer **64**. The upper connection link **72** may be spaced apart from the side **612** and the lower connection link **82** by the sub rotation shaft spacer **64**.

(114) The upper link assembly **70** may include the upper main link **71**, the upper connection link **72**, a first upper sub link **73**, a second upper sub link **74**, and an upper door bracket **75**. The lower link assembly **80** may include the lower main link **81**, the lower connection link **82**, a first lower sub link **83**, a second lower sub link **84**, and a lower door bracket **85**. The upper and lower door brackets **75** and **85** may alternatively be referred to as second and third brackets.

(115) The upper link assembly **70** and the lower link assembly **80** may be configured as a combination of a plurality of links and may have a same or similar coupling structure so as to have the same or similar rotation trajectory. The upper link assembly **70** and the lower link assembly **80** may be configured to have the same or similar rotational trajectory as the upper hinge **40** and the lower hinge **50**, and the upper door **20** and the lower door **30** may smoothly rotate.

(116) Referring to FIGS. **18-21**, the first upper sub link **73** and the second upper sub link **74** may be rotatably connected to each other between the upper main link **71** and the door bracket **75**. The upper connection link **72** may connect the second upper sub link **74** to the hinge bracket **61**.

(117) The upper main link **71** may be made of a plate-shaped metal material and be bent several times to accommodate at least portion of the upper connection link **72**, the first upper sub link **73**, and the second upper sub link **74** therein. The upper main link **71** may have a shape that does not interfere with the hinge bracket **61** and the upper door **20** during the rotation and may be provided so as not to interfere with the door bracket **75**. The hinge bracket **61**, the upper main link **71**, and the door bracket **75** may be provided so as not to interfere with each other during the folding or unfolding.

(118) The upper main link **71** may include a top surface **711**, a bottom surface **712**, and a connection surface or side wall **713** connecting the top surface **711** to the bottom surface **712**. The top surface **711** and the bottom surface **712** may be vertically bent from upper and lower ends of the connection surface **713**, respectively, and may extend in the same direction. The top surface **711** and the bottom surface **712** may have the same outer shape and may be provided to face each other. A predetermined or prescribed inner space **710** may be defined by the top surface **711**, the bottom surface **712**, and the connection surface **713**.

(119) The upper main hole **714** through which the main rotation shaft **62** passes may be provided in one end of each of the top surface **711** and the bottom surface **712**. The forming portion **712a** protruding downward may be further provided at one end of the bottom surface **712**, and the upper main hole **714** may be provided in the forming portion **712a** of the bottom surface **712**. The forming portion **712a** may have a shape protruding downward to be in contact with a top surface of the lower main link **81**. The bottom surface of the upper main link **71** and the top surface of the lower main link **81** may be spaced apart from each other by the forming portion **712a**.

(120) The top surface **711** and the bottom surface **712** of the upper main link **71** have shaft holes **715** and **716** in which the ends of the first upper sub link **73** and the second upper sub link **74** are axially coupled to each other, respectively. Shafts **717** and **718** may pass through the shaft holes **715** and **716**, and the first upper sub link **73** and the second upper sub link **74** may be rotatably coupled to each other. A spacer **719** may be provided on each of the shafts **717** and **718**. Each of the first upper sub link **73** and the second upper sub link **74** may be provided at an appropriate or predetermined height between the top and bottom surfaces of the upper main link **71** by the spacer **719** to secure smooth rotation without interfering with the upper main link **71** and the upper connection link **72**.

(121) The spacer **719** may have a cylindrical shape, and a hole may be provided in a center of the spacer **719** so that the shafts **717** and **718** pass in the vertical direction. A spacer groove **719a** guiding an insertion of the spacer **719** may be provided in a circumferential surface of the spacer **719**. When an end of each of the first upper sub link **73** and the second upper sub link **74** is inserted onto the spacer groove **719a**, the shafts **717** and **718** may pass through the upper main link **71**, the spacer **719**, and the first upper sub link **73** or the second upper sub link **74**.

(122) The first upper sub link **73** and the second upper sub link **74** may be coupled to each other in a state of being rotatable inside the upper main link **71**. A vertical height of each of the first upper sub link **73** and the second upper sub link **74** may be maintained in the inner space **710** of the upper main link **71** by the spacer **719a** and be rotatable without interfering with other components constituting the upper link assembly **70**.

(123) The top surface **711** and the bottom surface **712** of the upper main link **71** may extend to have

a predetermined or prescribed width. The top surface **711** and the bottom surface **712** of the upper main link **71** may be provided in a stepped shape, and strength may be reinforced to reduce a deformation of the upper main link **71** even if a load is applied.

(124) The top surface **711** and the bottom surface **712** of the upper main link **71** may include a first section **711b** in which the upper main hole **714** and the shaft holes **715** and **716** are formed and a second section **711c** extending forward from the first section **711b** and bent to be stepped in the vertical direction. A boundary between the first section **711b** and the second section **711c** may be stepped to provide a bent line or alternatively a straight line, thereby reinforcing the strength of the top surface **711** and the bottom surface **712** of the upper main link **71**. An interval between the top surface **711** and the bottom surface **712** of the upper main link **71** in the second section **711c** may be greater than that in the first section **711c**.

(125) A main tilted portion or section **711a** may be provided on one end of each of the top surface **711** and the bottom surface **712** of the upper main link **71**. When the upper link assembly **70** is fully folded, the main tilted portion **711a** may be provided to be parallel to a bracket tilted portion **751a** of the door bracket **75** so as not to interfere with the door bracket **75**. Each of the top surface **711** and the bottom surface **712** of the upper main link **71** may have a shape that does not interfere with the door bracket **75** during the operation of the upper link assembly **70**.

(126) A thickness of the upper main link **71** may be determined by the connection surface **713**, and the connection surface **713** may have a minimum or predetermined width configured to receive and accommodate the upper connection link **72**, the first upper sub link **73**, and the second upper sub link **74**.

(127) A connection surface opening **713a** that is opened at a position facing the shaft **715** may be provided in the connection surface **713**. The connection surface opening **713a** may be opened so that an end of the second upper sub link **74** does not interfere with the connection surface **713** during the rotation of the second upper sub link **74**. The structure of each of the upper link assembly **70** and the upper main link **71** may be more compact by the connection surface opening **713a**.

(128) The first upper sub link **73** may be axially coupled to the end of the upper main link **71** by the shaft **718**. The first upper sub link **73** may be provided in a metal plate shape and may be supported by the spacer **719** in an inner space of the upper main link **71**.

(129) The first upper sub link **73** may extend to be connected to one side of the door bracket **75** and may be rotatably coupled by the door bracket **75** and the shaft **754**. The first upper sub link **73** may have shaft holes **731** and **732** in both ends (i.e., first and second ends) thereof. Rotation shafts **718** and **754** may be coupled to the shaft holes **731** and **732**, respectively, so as to be rotatably coupled to the upper main link **71** and the door bracket **75**, respectively. A washer **756** may be provided on the shaft **754** connecting the first upper sub link **73** to facilitate a smooth rotation of the first upper sub link **73**.

(130) An outer portion of a circumferential or outer surface of the first upper sub link **73** may be provided in a linear shape. When the upper link assembly **70** is folded, the outer portion of the circumferential surface of the first upper sub link **73** may correspond to an outer end of the upper main link **71**, which may have a straight-line shape.

(131) A first tilted portion **734a** and a second tilted portion **734b** may be provided inside the first upper sub link **743** based on a sub recessed portion **733**. The first tilted portion **734a** may extend from the sub recessed portion **733** to a front end and may be in contact with the tilted portion **747** of the second upper sub link **74** when the upper link assembly **70** is fully folded. The second tilted portion **734b** extends from the sub recessed portion **733** to a rear end and is in contact with the tilted portion **747** of the second upper sub link **74** when the upper link assembly **70** is fully unfolded. A slope of the first tilted portion **734a** and a slope of the second tilted portion **734b** may be different from each other, and a slope of the second tilted portion **734b** may be greater than that of the first tilted portion **734a**.

(132) A sub recessed portion **733** may be provided in an inner partial section of the outer surface of the first upper sub link **73** at a side facing the second upper sub link **74**. The sub recessed portion **733** may have a shape corresponding to that of an end of the second upper sub link **74**, and when the upper link assembly **70** is fully unfolded, the sub recessed portion **733** and an end of the second upper sub link **74** may be fitted into each other.

(133) When the upper link assembly **70** is fully unfolded, the first upper sub link **73** and the second upper sub link **74** may move to complete the rotation of the upper link assembly **70**. Ends of the first upper sub link **73** and the second upper sub link **74** may have shapes corresponding to each other so as not to interfere with each other until the upper door **20** is fully opened. When the upper door **20** is fully opened, the upper link assembly **70** may no longer rotate due to being coupled and engaged.

(134) The second upper sub link **74** may have a plate shape made of a metal material, and one end of the second upper sub link **74** may be rotatably mounted to the upper main link **71** by the shaft **717**. The end of the second upper sub link **74** may be rotatably mounted between the first upper sub link **73** and the main rotation shaft **62**.

(135) The second upper sub link **74** may extend to be connected to the door bracket **75** and may be rotatably coupled to the door bracket **75** by the shaft **755**. The second upper sub link **74** may have shaft holes **741** and **742** in both (i.e., first and second) ends thereof, and shafts **717** and **755** may be inserted through the shaft holes **731** and **732**, respectively so as to be rotatably coupled to the upper main link **71** and the door bracket **75**, respectively. A washer **757** may be provided on the shaft **755** connecting the second upper sub link **74** to facilitate a smooth rotation of the second upper sub link **74**.

(136) A tilted portion **747** may be provided at one side of a circumference or outer side of the second upper sub link **74**, and when the upper link assembly **70** is fully folded, the tilted portion **747** may have a slope corresponding to that of the first tilted portion **734a** of the first upper sub link **73** to be in contact with the first tilted portion **734a**.

(137) A sub insertion portion **745** having a shape corresponding to that of the sub recessed portion **733** may be provided in one end of the second upper sub link **74**. The sub insertion portion **745** may be coupled to be engaged with the sub recessed portion **733** when the upper link assembly **70** is fully opened, and the upper link assembly **70** may not further rotate.

(138) Each of ends of the first upper sub link **73** and the second upper sub link **74** may be connected to the door bracket **75**. The door bracket **75** may be configured to support the upper door **20** at the lower side and may connect the upper link assembly **70** to the upper door **20**.

(139) The door bracket **75** may be made of a metal material and include a plate-shaped upper plate **751** and an upper shaft **752**. The upper plate **751** may have a plate shape, and a plate hole **753** through which an end of each of the first upper sub link **73** and the second upper sub link **74** is axially coupled may be provided. One plate hole **753** may be provided in each of both sides, and the first upper sub link **73** and the second upper sub link **74** may be rotatably connected to each other by the shafts **754** and **755**.

(140) A portion of a circumference or outer surface of the upper plate **751** may have a bracket tilted portion **751a** corresponding to the main tilted portion **711a** of the upper main link **71**. When the upper link assembly **70** is fully folded, the main tilted portion **711a** of the upper main link **71** and the corresponding tilted portion **751a** may be provided to face each other so as not to interfere with each other. An opposite side of the bracket tilted portion **751a** on the circumference of the upper plate **751** may have a straight line shape and be provided in parallel to or along a same extension line as the end of the upper main link **71** when the upper link assembly **70** is folded.

(141) The upper shaft **752** may extend upward from the upper plate **751** and include a pair of shafts are spaced apart from each other. The upper shaft **752** may be inserted into the bottom surface of the upper door **20**, and the upper door **20** may be coupled to rotate together with the upper link assembly **70**. An end of each of the first upper sub link **73** and the second upper sub link **74** may be

coupled to the upper plate **751** at the bottom surface of the upper plate **751** so as not to interfere with the upper shaft **752**.

(142) The shafts **754** and **755** connecting the upper plate **751**, the first upper sub link **73**, and the second upper sub link **74** to each other may be provided with washers **756**, **756a**, **757**, and **757a**. The first upper sub link **73** and the second upper sub link **74** may be stably axially coupled to the upper plate **751**.

(143) A sub connection hole **743** may be further provided in the second upper sub link **74** between the shaft holes **741** and **742** at both ends of the second upper sub link **74**. The second upper sub link **74** may be connected to the upper connection link **72**. The shaft **744** passing through the sub connection hole **743** of the second upper sub link **74** may be coupled to pass through the connection hole **726** of the upper connection link **72**, and the second upper sub link **74** and the upper connection link **72** may be rotatably coupled to each other.

(144) The upper connection link **72** may be provided in an inner space of the upper main link **71** to prevent the first upper sub link **73** and the second upper sub link **74** from drooping downward and assist the rotation of the second upper sub link **74**. One end of the upper connection link **72** may be rotatably connected to the hinge bracket **61** by the sub rotation shaft **63** to extend to an end of the upper main link **71**. The upper connection link **72** may extend to pass below the first upper sub link **73** and the second upper sub link **74**. The upper connection link **72** may be rotatably provided inside the upper main link **71** to support the first upper sub link **73** and the second upper sub link **74** at the lower side.

(145) A sub rotation shaft hole **721** through which the sub rotation shaft **63** passes may be provided in one end of the upper connection link **72**. The upper connection link **72** may be mounted on the hinge bracket **61** to rotate based on the sub rotation shaft **63**.

(146) A protrusion **722** may be provided on one end of the upper connection link **72** in which the sub rotation shaft hole **721** is provided. When the upper link assembly **70** is fully folded, the protrusion **722** may be in contact with an inner surface of the shielding portion **614** to restrain the upper link assembly **70** so that the upper link assembly **70** is not folded any more.

(147) A front recessed or bent portion **723** may be provided at one side of a circumferential or outer surface of the upper connection link **72** which faces a front side. The front recessed portion **723** may be recessed so as not to interfere with the shaft **754** of the first upper sub link **73** when the upper link assembly **70** is fully folded.

(148) A first rear recessed or bent portion **724** and a second rear recessed or bent portion **725** may be provided at one side of the circumferential surface of the upper connection link **72** which faces a rear side. The first rear recessed portion **724** and the second rear recessed portion **725** may be recessed so as not to interfere with the shaft **717** coupled to the second upper sub link **74** according to the operation of the upper link assembly **70**.

(149) For example, when the upper link assembly **70** is fully unfolded, the shaft **717** coupled to the second upper sub link **74** may be provided at the first rear recessed portion **724**, and in the state in which the upper link assembly **70** is fully folded, the shaft **717** coupled to the second upper sub link **74** may be provided at the second rear recessed portion **725**.

(150) A connection hole **726** connected to the second upper sub link **74** may be further provided in the upper connection link **72**. The shaft **744** passing through the sub connection hole **743** of the second upper sub link **74** may be coupled to pass through the connection hole **726**. The second upper sub link **74** may rotate in a state in which the upper main link **71** and the upper connection link **72** are axially coupled to each other. The washer **746** penetrated by the shaft **744** may be provided above the upper connection link **72**.

(151) The connection hole **726** may be provided at an approximately intermediate point of the upper connection link **72**. The upper connection link **72** may include a connection portion or section **727** and a support portion or section **728** which extend to left and right sides based on the connection hole **726**. The connection hole **426** and the sub rotation shaft hole **721** may be provided

in the connection portion **727**, and each of the upper connection link **72** and the second upper sub link **74** may be rotatably coupled. The support portion **728** may be provided to extend to the first upper sub link **73** through at least the second upper sub link **74** and be provided in the inner space **710** of the upper main link **71** under the first upper sub link **73** and the second upper sub link **74**.

(152) A support member **729** may be further provided at one end of the support portion **728**. The support member **729** may be mounted on the support portion **728** and be made of a plastic material. For example, the support member **729** may be made of an engineering plastic material or a polyoxymethylene (POM) material having wear resistance and lubricity.

(153) A support member mounting portion **728a** on which the support member **729** is mounted may be provided to be stepped or recessed in the support portion **728**. A protrusion **729a** may be provided on a bottom surface of the support member **729**, and a groove **728b** may be provided in the support member mounting portion **728a**. The support member **729** may be fixed and mounted to the support portion **728**.

(154) The support member **729** may protrude more than a top surface of the upper connection link **72** when mounted on the support portion **728**. The support member **729** may be in contact with a bottom surface of the first upper sub link **73** while the first upper sub link **73** rotates.

(155) The links **72**, **73**, and **74** may be provided in the inner space **710** of the upper main link **71** and may have a slim vertical thickness to reduce a total thickness of the multi-joint link hinge **60**. During the operation of the upper link assembly **70**, the first upper sub link **73** or the second upper sub link **74** may structurally droop downward. The support member **729** of the upper connection link **72** may support the first upper sub link **73** or the second upper sub link **74** to prevent the first upper sub link **73** or the second upper sub link **74** from drooping downward, preventing the first upper sub link **73** or the second upper sub link **74** from interfering with other components.

(156) When a heavy object is accommodated in the upper door **20** or an excessive downward load is applied, the upper link assembly **70** may partially droop. While the upper door **20** is opened and closed, the first upper sub link **73** and the second upper sub link **74** may be unfolded to be more susceptible to drooping.

(157) Even if the first upper sub link **73** or the second upper sub link **74** droops, the first upper sub link **73** and the second upper sub link **74** may be supported by the support portion **725** at the lower side. The first upper sub link **73** may be in contact with the support member **729** if downward drooping occurs during the rotational operation. The rotation of the first upper sub link **73** may be more smoothly performed. When the first upper sub link **73** is supported by the protruding support member **729**, the second upper sub link **74** may be maintained to be spaced apart from the upper connection link **72**. The second upper sub link **74** may rotate smoothly without interfering with the upper connection link **72**.

(158) Referring to FIGS. **22-23**, the lower link assembly **80** may have the same or similar planar shape as the upper link assembly **70** as a whole and may have the same or similar structure and shape as the upper link assembly **70** when viewed from an upper side even in the state of being folded or unfolded. The lower link assembly **80** may include the lower main link **81**, the first lower sub link **83**, the second lower sub link **84**, and a lower door bracket **85**. The first lower sub link **83** and the second lower sub link **84** may be rotatably connected to each other between the lower main link **81** and the door bracket **85**. The lower link assembly **80** may further include the lower connection link **82** connecting the second lower sub link **84** to the hinge bracket **61**.

(159) The lower main link **81** may be made of a plate-shaped metal material. Like the planar shape of the upper main link **71**, the lower main link **81** may have a shape that does not interfere with the hinge bracket **61**, the lower door **30**, and the lower door bracket **85** during the rotation. The hinge bracket **61**, the lower main link **81**, and the lower door bracket **85** may be configured so as not to interfere with each other during the folding or unfolding.

(160) A lower main hole **811** through which the main rotation shaft **62** passes may be provided in one end of the lower main link **81**. A spacer **817** through which the main rotation shaft **62** passes

may be further provided below the lower main hole **811**, and the lower main link **81** may be maintained to be spaced apart from the bottom surface of the hinge bracket **61**.

(161) Shaft holes **812** and **813** in which ends of the first lower sub link **83** and the second lower sub link **84** are axially coupled, respectively, may be provided in the lower main link **81**. The shafts **814** and **815** may pass through the shaft holes **812** and **813**, and the first lower sub link **83** and the second lower sub link **84** may be rotatably coupled to each other. At least one washer **816**, **816a**, **817**, and **817a** may be provided on the shafts **814** and **815**. An interval space between the lower main link **81**, the first lower sub link **83**, and the second lower sub link **84** may be constantly maintained by the washers **816**, **816a**, **817**, and **817a** to facilitate a smooth rotational operation.

(162) A main tilted portion **711a** may be provided on one end of the lower main link **81**. When the lower link assembly **80** is fully folded, the main tilted portion **711a** may be provided to be parallel to the bracket tilted portion **751a** of the lower door bracket **85** so as not to interfere with the lower door bracket **85**.

(163) The lower main link **81** may have a plate shape. A thickness of the lower main link **81** may be thinner than that of the upper main link **71**. A thickness of the entire lower link assembly **80** may be significantly less than that of the upper link assembly **70**. The lower link assembly **80** may simply fix the upper end of the lower door **30** so that a substantial load of the lower door **30** is supported by the lower hinge **50**. The thickness of the upper link assembly **70** may be reduced or minimized to accommodate the thickness of the lower link assembly **80**, and the multi-joint link hinge **60** may have a reduced thickness.

(164) A pair of supporters **86** may be provided on a top surface of the lower main link **81**. The supporter **86** may support the upper link assembly **70** so as not to be caught and restricted by the lower link assembly **80** even if downward drooping occurs during rotation of the upper link assembly **70**. The supporter **86** may be made of engineering plastic or POM material having abrasion resistance and lubricity. When the upper link assembly **70** is in contact with the supporter **86**, the upper link assembly **70** may not be caught, but smoothly rotate.

(165) The first lower sub link **83** may have a metal plate shape and may be axially coupled to an end of the lower main link **81** by the shaft **814**. The first lower sub link **83** may extend to be connected to one side of the lower door bracket **85** and may be rotatably coupled by the lower door bracket **85** and the shaft **855**. The first lower sub link **83** may have shaft holes **831** and **832** provided in both ends thereof, and shafts **814** and **855** may be inserted through the shaft holes **831** and **832**, respectively, to be rotatably coupled to the lower main link **81** and the lower door bracket **85**, respectively. Washers **816**, **816a**, and **857** may be provided on the shaft **814** and **855** to which the first lower sub link **83** is axially coupled to allow the first lower sub link **83** to smoothly rotate.

(166) A portion of an outer portion of a circumferential or outer surface of the first lower sub link **83** may be provided in a linear shape. In a state in which the lower link assembly **80** is folded, an outer portion of the circumferential surface of the first lower sub link **83** may have a straight-line shape to correspond to an outer end of the lower main link **81**. A sub recessed portion **833** may be provided in an inner partial section of the circumferential surface of the first lower sub link **83** at a side facing the second lower sub link **84**.

(167) A first tilted portion **834a** and a second tilted portion **834b** may be provided inside the first lower sub link **84** based on a sub recessed portion **833**. The first tilted portion **834a** may extend from the sub recessed portion **833** to a front end and be in contact with the tilted portion **847** of the second lower sub link **84** when the lower link assembly **80** is fully folded. The second tilted portion **834b** may extend from the sub recessed portion **833** to a rear end and be in contact with the tilted portion **847** of the second lower sub link **84** when the lower link assembly **80** is fully unfolded. A slope of the first tilted portion **834a** and a slope of the second tilted portion **834b** may be different from each other, and a slope of the second tilted portion **834b** may be greater than that of the first tilted portion **834a**.

(168) The sub recessed portion **833** may have a shape corresponding to that of an end of the second

lower sub link **84**, and when the lower link assembly **80** is fully unfolded, the sub recessed portion **833** and an end of the second lower sub link **84** may be fitted into each other. Even if the lower link assembly **80** is fully unfolded, the first lower sub link **83** and the second lower sub link **84** may interfere with each other to complete the rotation of the lower link assembly **80**. Ends of the first lower sub link **83** and the second lower sub link **84** may have shapes corresponding to each other so as not to interfere with each other until the lower door **30** is fully opened. When the lower door **30** is fully opened, the lower link assembly **80** no longer rotates due to being coupled and engaged. (169) The second lower sub link **84** may have a plate shape made of a metal material, and one end of the second lower sub link **74** may be rotatably mounted to the upper main link **81** by the shaft **815**. One end of the second lower sub link **84** may be rotatably mounted between the shaft **814** to which the first lower sub link **83** is coupled and the main rotation shaft **62**.

(170) The second lower sub link **84** may extend to be connected to the lower door bracket **85** and may be rotatably coupled by the lower door bracket **85** and the shaft **854**. The second lower sub link **84** may have shaft holes **841** and **842** in both ends thereof, and shafts **815** and **854** may be coupled to the shaft holes **841** and **842** so as to be rotatably coupled to the lower main link **81** and the lower door bracket **85**, respectively. Washers **817**, **817a**, and **856** may be provided on the shaft **815** and **854** to which the second lower sub link **84** is axially coupled to allow the second lower sub link **84** to smoothly rotate.

(171) A sub insertion portion or tab **845** having a shape corresponding to that of the sub recessed portion **733** may be provided at an end of the second lower sub link **84** through which the shaft **854** passes. The sub insertion portion **845** may be coupled to be engaged with the sub recessed portion **833** when the lower link assembly **80** is fully opened so that the lower link assembly **80** does not further rotate.

(172) Each of the ends of the first lower sub link **83** and the second lower sub link **84** may be connected to the lower door bracket **85**. The lower door bracket **85** may be connected to the top surface of the lower door **30** to connect the lower link assembly **80** to the lower door **30**.

(173) The lower door bracket **85** may be made of a metal material and include a plate-shaped lower plate **851** and a lower shaft **852**. The lower plate **851** may have a plate shape, and a plate hole **853** through which an end of each of the first lower sub link **83** and the second lower sub link **84** is axially coupled may be provided. One plate hole **853** may be provided in each of both sides, and the first lower sub link **83** and the second lower sub link **84** may be rotatably connected to each other by the shafts **854** and **855**.

(174) A portion of a circumference or outer surface of the lower plate **851** may have a tilted portion **851a** corresponding to the tilted portion **811a** of the lower main link **81**. When the lower link assembly **80** is fully folded, the tilted portion **811a** of the lower main link **81** and the corresponding tilted portion **851a** may be provided to face each other so as not to interfere with each other. An opposite side of the tilted portion **851a** on the circumference of the lower plate **851** may have a straight line shape and be provided in parallel to or in the same extension line as the end of the lower main link **81** when the lower link assembly **80** is folded.

(175) The lower shaft **852** may extend downward from the lower plate **851** and be provided as a pair of shafts spaced apart from each other. The lower shaft **852** may be inserted into the bottom surface of the lower door **20**, and the lower door **30** may be coupled to rotate together with the lower link assembly **80**. An end of each of the first lower sub link **83** and the second lower sub link **84** may be coupled to the lower plate **851** at the top surface of the lower plate **851** so as not to interfere with the lower shaft **852**. The shafts **854** and **855** connecting the lower plate **851**, the first lower sub link **83**, and the second lower sub link **84** to each other may be provided with at least one or more washers **856** and **857**, and the first lower sub link **83** and the second lower sub link **84** may be stably axially coupled to the lower plate **851**.

(176) A sub connection hole **843** may be provided in the second lower sub link **84** between the shaft holes **841** and **842** at each of both ends of the second lower sub link **84**. The shaft **844** may

pass through the sub connection hole **843** of the second lower sub link **84** and may be coupled to pass through the connection hole **826** of the lower connection link **82**. The second lower sub link **84** and the lower connection link **82** may be rotatably coupled to each other.

(177) The lower connection link **82** may be provided under the first lower sub link **83** and the second lower sub link **84** to prevent the first lower sub link **83** and the second lower sub link **84** from drooping downward and assist the rotation of the second lower sub link **84**. One end of the lower connection link **82** may be rotatably connected to the hinge bracket **61** by the sub rotation shaft **63** to extend to an end of the lower main link **81**. The lower connection link **82** may extend to pass below the first lower sub link **83** and the second lower sub link **84**. The lower connection link **82** may be rotatably provided inside the lower main link **81** to support the first lower sub link **83** and the second lower sub link **84** at the lower side.

(178) A sub rotation shaft hole **821** through which the sub rotation shaft **63** passes may be defined in one end of the lower connection link **82**. The lower connection link **82** may be mounted on the hinge bracket **61** to rotate based on the sub rotation shaft **63**.

(179) A front recessed portion **822** may be provided at one side of a circumferential or outer surface of the lower connection link **82** which faces a front side. The front recessed portion **822** may be recessed so as not to interfere with the shaft **855** of the first lower sub link **83** when the lower link assembly **80** is fully folded.

(180) A rear recessed portion **823** may be provided at one side of a circumferential or outer surface of the lower connection link **82** which faces the rear side. The rear recessed portion **823** may be recessed so as not to interfere with the shaft **814** coupled to the first lower sub link **83** according to the operation of the lower link assembly **80**. For example, a shaft **814** coupled to the first lower sub link **83** may be provided in the rear recessed portion **823** when the lower link assembly **80** is fully unfolded.

(181) A connection hole **826** connected to the second lower sub link **84** may be further provided in the lower connection link **82**. The shaft **844** passing through the sub connection hole **743** of the second lower sub link **84** may be coupled to pass through the connection hole **826**. The second upper sub link **74** may rotate when the upper main link **71** and the upper connection link **72** are axially coupled to each other. The connection hole **826** may be provided at an approximately intermediate point of the upper connection link **72**. The washers **846** and **846a** penetrated by the shaft **844** may be provided above and below the lower connection link **82**.

(182) A bent portion **824** may be provided on the lower connection link **82**. The lower connection link **82** may include a first connection portion **825** and a second connection portion **827** based on the bent portion **824**. The first connection portion **825** may be provided higher than the second connection portion **827** by the bent portion **824**, and the lower connection link **82** may be stepped. A difference in height between the first connection portion **825** and the second connection portion **827** may be greater than a thickness of each of the first lower sub link **83** and the second lower sub link **84**. A space may be defined between an upper side of the second connection portion **827** and a lower side of the lower main link **81** so that the first lower sub link **83** and the second lower sub link **84** may rotate smoothly.

(183) A sub rotation shaft hole **821** through which the sub rotation shaft **63** passes may be provided in an end of the first connection portion **825**. The lower connection link **82** may rotate about the sub rotation shaft **63** in the hinge bracket **61**. A connection hole **826** may be provided in the second connection portion **827**, and the second upper sub link **74** may have a rotatable coupling structure. The lower connection link **82** and the second lower sub link **84** may be rotatable with each other around the shaft **844**.

(184) The second connection portion **827** may extend to the first lower sub link **83** via the second lower sub link **84** to support the first lower sub link **83** under the first lower sub link **83** and the second lower sub link **83**. A support member **829** may be further provided at one end of the second connection portion **827**. The support member **829** may be mounted on an end of the second

connection portion **828** and be made of a plastic material (e.g., an engineering plastic material or a POM material having wear resistance and lubricity).

(185) A support member mounting portion **828a** on which the support member **829** is mounted may be provided to be stepped or recessed in the second connection portion **827**. A protrusion **829a** may be provided on a bottom surface of the support member **829**, and a groove **828b** may be defined in the support member mounting portion **828a**. The protrusion **829a** and the groove **828b** may be coupled to each other so that the support member **829** is fixed to the support member mounting portion **828a**.

(186) The support member **829** may protrude more than a top surface of the upper connection link **72** when mounted on the second connection portion **827**. The support member **829** may be in contact with a bottom surface of the first lower sub link **83** while the first lower sub link **83** rotates.

(187) The lower link assembly **80** may have a plate-like structure having a thin thickness to reduce a total thickness of the multi-joint link hinge **60**. During the operation of the lower link assembly **80**, the first lower sub link **83** or the second lower sub link **84** may structurally droop downward. The support member **829** of the lower connection link **82** may support the first lower sub link **83** or the second lower sub link **84** to prevent the first lower sub link **83** or the second lower sub link **84** from drooping downward, thereby preventing the first lower sub link **73** or the second lower sub link **74** from interfering with other components.

(188) When a heavy object is provided in the lower door **30** or an excessive downward load is applied, the lower link assembly **80** may partially droop. While the lower door **30** is opened and closed, the first lower sub link **83** and the second lower sub link **84** may be unfolded to be more susceptible to the drooping.

(189) Even if the first lower sub link **83** and the second lower sub link **84** droop, the first lower sub link and the second lower sub link may be supported by the second connection portion **827** at the lower side. The first lower sub link **83** may be in contact with the support member **829** if downward drooping occurs during rotation to facilitate a smooth rotation of the first lower sub link **83**. When the first lower sub link **83** is supported by the protruding support member **829**, the second lower sub link **84** may be naturally maintained in a state of being spaced apart from the lower connection link **72**. The second lower sub link **84** may rotate smoothly without interfering with the lower connection link **72**.

(190) Referring to FIGS. **25-26**, the multi-joint link hinge **60** may be provided between the upper door **20** and the lower door **30** and may be mounted on the front surface of the cabinet **10**. A total thickness **T** of the multi-joint link hinge **60** may correspond to a vertical width of the hinge bracket **61**. The thickness **T1** of the upper link assembly **70** may be greater than a thickness **T2** of the lower link assembly **80**. Since the upper link assembly **70** and the lower link assembly **80** are provided in the receiving space **610** inside the hinge bracket **61**, the sum of the thickness **T1** of the upper link assembly **70** and the thickness **T2** of the lower link assembly **80** may be equal to or slightly less than the thickness **T** of the hinge bracket **61**.

(191) The multi-joint link hinge **60** may be between the bottom surface of the upper door **20** and the top surface of the lower door **30**. When viewed from the front side, at least a portion of the multi-joint link hinge **60** may be covered.

(192) An upper decor or frame **22** may be provided on the bottom surface of the upper door **20**. The upper decor **22** may define the bottom surface of the upper door **20**, and an upper link mounting portion **221** may be provided on one end of the upper decor **22**. The upper link mounting portion **221** may be recessed upward, and the upper shaft **752** of the upper link assembly **70** may be inserted into the upper link mounting portion **221**. The door bracket **45** may be coupled to the upper link mounting portion **221**, and the upper link assembly **70** may be mounted to support the upper door **20** at the lower side and rotate by the upper link assembly **70**.

(193) A lower decor **32** may be provided on the bottom surface of the lower door **30**. The lower decor **32** may define a top surface of the lower door **30**, and the lower link mounting portion **321**

may be provided on one end of the lower decor **32**. The lower link mounting portion **321** may be recessed downward, and the lower shaft **852** of the lower link assembly **80** may be inserted into the upper link mounting portion **221**. The door bracket **55** may be coupled to the lower link mounting portion **321**, and the lower door **30** may rotate by the lower link assembly **80**.

(194) A decor recessed portion or recess **322** may be provided in the lower decor **32**. The decor recessed portion **322** may be recessed downward, and a lower link mounting portion **321** may be provided at the decor recessed portion **322**.

(195) The decor recessed portion **322** may be recessed by a predetermined depth **H1** and be provided under the multi-joint link hinge **60**. When viewed from the front side, the lower portion of the multi-joint link hinge **60** may be covered by the lower door **30**. A height of the upper end of the lower door **30** may be less than that of the lower end of the upper link assembly **70**. When the upper link assembly **70** operates, the upper link assembly **70** may not interfere with the upper end of the lower door **30**.

(196) A distance **H1** between the lower end of the upper door **20** and the upper end of the lower door **30** may be less than the thickness **T** of the multi-joint link hinge **60** and greater than the thickness **T1** of the upper link assembly **70**. While the upper link assembly **70** may operate smoothly, an exposure of the multi-joint link hinge **60** may be reduced or minimized. The multi-joint link hinge **60** may allow the upper door **20** and the lower door **30** to reduce or minimize the exposure of the multi-joint link hinge **60** between the upper door **20** and the lower door **30** while allowing the upper door **20** and the lower door **30** to be smoothly opened and/or closed. A gap between the upper door **20** and the lower door **30** may be reduced to improve an outer appearance when the upper door **20** and the lower door **30** are closed.

(197) The multi-joint link hinge **60** may be mounted on a front surface of a barrier plate **112** defining the front surface of the barrier **11**. The multi-joint link hinge **60** may be provided at an intermediate position with respect to a vertical height of the barrier **11**.

(198) The barrier **11** may include a barrier case **111** in which an insulating material is provided and the barrier plate **112**. The barrier plate **112** may be mounted in a recessed space in front surface of the barrier case **111** to define the front surface of the barrier **11**. A pair of heating members **113** may be provided on a rear surface of the barrier plate **112**. The heating members **113** may be provided along the barrier **11** and be provided at positions corresponding to the upper door gasket **21** and the lower door gasket **31** on the rear surfaces of the upper door **20** and the lower door **30**, respectively. Magnets **211** and **311** may be embedded in the upper door gasket **21** and the lower door gasket **31**, respectively. The upper door gasket **21** and the lower door gasket **31** may be in close contact with the front surface of the barrier plate **112** made of a metal material.

(199) Each of the heating members **113** may be provided on the rear surface of the barrier plate **112** corresponding to the position of each of the magnets **211** and **311** to extend in a horizontal direction along the barrier plate **112**. The heating members **113** may be arranged vertically and may be spaced a predetermined or prescribed distance **H3** from each other. The multi-joint link hinge **60** may be provided between the heating members **113**. Each of the thickness of the multi-joint link hinge **60** and the vertical width of the hinge bracket **61** may be less than the distance **H3** between the heating members **113**.

(200) Even if the hinge bracket **61** is screwed on the barrier **11**, the heating member **113** may be prevented from being damaged. Since ends of the upper door gasket **21** and the lower door gasket **31** may be provided at positions corresponding to the heating members **113**, respectively, the upper door gasket **21** and the lower door gasket **31** may be prevented from interfering or colliding with the multi-joint link hinge **60**.

(201) The multi-joint link hinge **60** may be provided between the upper door gasket **21** and the lower door gasket **31**, but may not be provided at a center between the upper door gasket **21** and the lower door gasket **31**. The multi-joint link hinge **60** may be provided farther from a lower end of the upper door gasket **21** than from an upper end of the lower door gasket **31**. The upper door

gasket **21** may droop downward by its own weight in the mounted state, and a distance between the multi-joint link hinge **60** and the upper door gasket **21** may be greater than that between the multi-joint link hinge **60** and the lower door gasket **31**. Even if the upper door gasket **21** droops or sags due to continuous, extended use, collision with the multi-joint link hinge **60** may be prevented.

(202) An exposure of the multi-joint link hinge **60** may be reduced or minimized, and when the upper link assembly **70** and the lower link assembly **80** operate, an interference or collision between the upper door gasket **21** and the lower door gasket **31** may be prevented to facilitate a smooth rotation of the upper door **20** and the lower door **30**.

(203) Referring to FIG. 27, when both the upper door **20** and the lower door **30** are closed, the multi-joint link hinge **60** may be in a compact state and have a substantially hexahedral shape. In this compact state, a space occupied by the multi-joint link hinge **60** may be reduced or minimized, and the multi-joint link hinge **60** may be effectively provided in a narrow space between the upper door **20** and the lower door **30**.

(204) The upper link assembly **70** and the lower link assembly **80** may be provided in an inner region of the hinge bracket **61** in the fully folded state. The first upper sub link **73**, the second upper sub link **74**, and the upper connection link **72** may be provided in the inner space of the upper main link **71** constituting the upper link assembly **70**.

(205) The upper door bracket **75** may be provided inside a space defined by the tilted portion **711a** of the top surface of the upper main link **71** and the front end of the hinge bracket **61**. The tilted portion **751a** of the rear surface of the door bracket **75** may be provided to face each of the tilted portion **711a** of the top surface of the upper main link **71** and the front end of the hinge bracket **61**, and the front surface of the door bracket **75** may be provided parallel to the front end of the upper main link **71**. In such an arrangement, the upper link assembly **70** may define a top surface shape of the multi-joint link hinge **60** together with the side **612** of the hinge bracket **61**.

(206) The lower link assembly **80** may be provided below the upper link assembly **70** in the fully folded state. The lower main link **81** may be provided below the upper main link **71**, and the first lower sub link **83** and the second sub lower link **84** may be provided below the lower main link **81** in the state of being fully folded.

(207) All the first lower sub link **83**, the second sub lower link **84**, and the lower door bracket **85** may be in the fully folded state and be maintained in the hexahedral shape as a whole without protruding to the outside of **60** like the upper link assembly **70**. A straight section of the front end of the upper main link **71**, a straight section of the front end of the first upper sub link **73**, and a straight section of the front end of the upper door bracket **75** may be provided in a same extension line or in a straight line adjacent to the extension line. Thus, the multi-joint link hinge **60** may have a compact and clean appearance in the folded state.

(208) When the user opens the upper door **20** or the lower door **30** from the compact state, the upper door **20** and the lower door **30** start to rotate, and the multi-joint link hinge **60** may operate. When the upper door **20** and the lower door **30** rotate, the upper hinge **40** connected to the upper end of the upper door **20** and the lower hinge **50** connected to the lower end of the lower door **30** may rotate along the same trajectory as the multi-joint link hinge **60**. The upper link assembly **70** may rotate in the same manner as the rotational trajectory of the upper hinge **40** by the guide of the upper hinge **40**, and the lower link assembly **80** may rotate in the same manner as the rotational trajectory of the lower hinge **50** by the guide of the lower hinge **50**.

(209) A structure and shape of the lower link assembly **80** may be the same as or similar to those of the upper link assembly **70**. The lower main link **81**, the first lower sub link **83**, the second lower sub link **84**, and the lower door bracket **85** may have the same or similar planar shape as the upper link assembly **70**. When the lower link assembly **80** is folded, the lower link assembly **80** may have the same outer appearance structure as the upper link assembly **70**.

(210) Referring to FIGS. 28-31, when the user opens the upper door **20**, the upper link assembly **70** starts to rotate. According to the rotation operation of the upper door **20**, the upper link assembly **70**

may operate from the fully folded state of FIG. 28 to the fully unfolded state of FIG. 31. The figures may demonstrate cases where an angle between the front surface of the upper door 20 and the front surface of the cabinet 10 is about 13° in FIG. 28, about 30° in FIG. 29, about 90° in FIG. 30, and about 130° in FIG. 31.

(211) According to the user's manipulation for the upper door 20, the upper hinge 40 and the upper link assembly 70 of the multi-joint link hinge 60 operate, and the upper door 20 may rotate together along the rotational trajectory of the upper link assembly 70 of the multi-joint link hinge 60 and then be opened. The upper link assembly 70 may operate and support the upper door 20 from the lower side. The upper link assembly 70 may be sequentially changed in state as illustrated in FIGS. 28 to 31 until the upper door 20 starts to rotate and then is fully opened.

(212) The upper main link 71 may rotate in a counterclockwise direction with respect to the main rotation shaft 62 together with the opening of the upper door 20, and the upper connection link 72 may start to rotate in a counterclockwise direction with respect to the sub rotation shaft 63. Ends of the first upper sub link 73 and the second upper sub link 74 connected to the upper main link 71 start to rotate in a clockwise direction. The upper door bracket 75 connected to the other ends of the first upper sub link 73 and the second upper sub link 74 may rotate in the counterclockwise direction.

(213) The upper main link 71, the first upper sub link 73, and the second upper sub link 74 may be provided inside the upper main link 71, and the upper main link 71, the first upper sub link 73, and the second upper sub link 74 may rotate while moving together with the upper main link 71. The upper connection link 72 may be provided under the first upper sub link 73 and the second upper sub link 74 to support the first upper sub link 73 and the second upper sub link 74, preventing drooping of the first upper sub link 73 and the second upper sub link 74.

(214) The first upper sub link 73 and the second upper sub link 74 may rotate while being spaced apart from each other, and the first tilted portion 734a of the first upper sub link 73 and the tilted portions 747 of the second upper sub link 74 may face each other. While the upper main link 71 and the upper connection link 72 continuously rotate according to an opening of the upper door 20, the first upper sub link 73 and the second upper sub link 74 also rotate together.

(215) The first tilted portion 734a of the first upper sub link 73 and the tilted portion 747 of the second upper sub link 74 may be spaced away from each other as illustrated in FIGS. 28 and 29. When the upper link assembly 70 is further unfolded, as illustrated in FIGS. 30 and 31, the second tilted portion 734b of the first upper sub link 73 and the tilted portion 747 of the second upper sub link 74 may be close to each other as illustrated in FIGS. 30 and 31. As illustrated in FIG. 31, the second tilted portion 734b of the first upper sub link 73 and the tilted portion 747 of the second upper sub link 74 may be in contact with each other in a fully opened state.

(216) In the fully opened state as shown in FIG. 31, the sub insertion portion 745 of the second upper sub link 74 may be inserted into the sub recessed portion 733 of the second upper sub link 74, and the tilted portion 747 of the second upper sub link 74 may be in contact with the second tilted portion 743b of the first upper sub link 73. The rotation of the first upper sub link 73 and the second upper sub link 74 may be restricted so as not to rotate any longer in the clockwise direction, and the rotation of the upper door bracket 75 in the counterclockwise rotation may be restricted. A further rotation of the upper door 20 may be prevented, and the upper door 20 may be considered to be completely or fully open.

(217) When the upper door 20 is fully opened, the upper link assembly 70 may be fully unfolded, and in this state, the upper main link 71 may droop or be deformed by the load of the upper door 20. Since the upper main link 71 may have a bent structure, the upper link assembly 70 may not be or be less deformed by the load of the upper door 20.

(218) The first upper sub link 73 and the second upper sub link 74 may be supported by the upper connection link 72 to prevent the first upper sub link 73 and the second upper sub link 74 from drooping. The user may perform an operation of closing the upper door 20, and while the upper

door **20** is closed, the upper link assembly **70** may operate in a reverse order of the above-described process.

(219) Referring to FIGS. **32-35**, when the user opens the lower door **30**, the lower link assembly **80** starts to rotate. According to a rotation of the lower door **30** during opening and closing, the lower link assembly **80** may operate from the fully folded state of FIG. **32** to the fully unfolded state of FIG. **35**. As an example, an angle between the front surface of the lower door **30** and the front surface of the cabinet **10** may be about 13° in FIG. **32**, about 30° in FIG. **33**, about 90° in FIG. **34**, and about 130° in FIG. **35**.

(220) According to the user's manipulation for the lower door **30**, the lower door **30** may rotate together along the rotational trajectory of the lower link assembly **80** of the multi-joint link hinge **60** and then be opened. The lower link assembly **80** may connect the lower door **30** to the cabinet **10** from the upper side. The lower link assembly **80** may be sequentially changed in state as illustrated in FIGS. **32** to **35** until the lower door **30** starts to rotate and then is fully opened.

(221) The lower main link **81** may rotate in a counterclockwise direction with respect to the main rotation shaft **62** together with the opening of the lower door **30**, and the lower connection link **82** may start to rotate in a counterclockwise direction with respect to the sub rotation shaft **63**. Ends of the first lower sub link **83** and the second lower sub link **84** connected to the lower main link **81** may start to rotate in a clockwise direction. The lower door bracket **85** connected to the other ends of the first lower sub link **83** and the second lower sub link **84** may rotate in the counterclockwise direction.

(222) When the lower main link **81**, the first lower sub link **83**, and the second lower sub link **84** are provided inside the lower main link **81**, the lower main link **81**, the first lower sub link **83**, and the second lower sub link **84** may rotate while moving together with the lower main link **81**. The lower connection link **82** may be provided under the first lower sub link **83** and the second lower sub link **84** to support the first lower sub link **83** and the second lower sub link **84**, thereby preventing the first lower sub link **83** and the second lower sub link **84** from drooping.

(223) The first lower sub link **83** and the second lower sub link **84** may rotate while being spaced apart from each other, and the first tilted portion **834a** of the first lower sub link **83** and the tilted portions **847** of the second lower sub link **84** may face each other. While the lower main link **81** and the lower connection link **82** continuously rotate according to an opening of the lower door **30**, the first lower sub link **83** and the second lower sub link **84** may also rotate together.

(224) As illustrated in FIGS. **32-33**, the first tilted portion **834a** of the first lower sub link **83** and the tilted portion **847** of the second lower sub link **84** may be spaced away from each. When the lower link assembly **80** is further unfolded, as illustrated in FIGS. **34** and **35**, the second tilted portion **834b** of the first lower sub link **83** and the tilted portion **847** of the second lower sub link **84** may be close to each other as illustrated in FIGS. **30** and **31**. Finally, the second tilted portion **834b** of the first lower sub link **83** and the tilted portion **847** of the second lower sub link **84** may be in contact with each other.

(225) FIG. **35** may indicate a state in which the lower door **30** is fully opened. The sub insertion portion **845** of the second lower sub link **84** may be inserted into the sub recessed portion **833** of the second lower sub link **84**, and the tilted portion **847** of the second lower sub link **84** is in contact with the second tilted portion **843b** of the first lower sub link **83**. A rotation of the first lower sub link **83** and the second lower sub link **84** may be restricted so as not to rotate any longer in the clockwise direction, and the rotation of the lower door bracket **85** in the counterclockwise rotation is restricted. A further rotation of the lower door **30** may be prevented, and the lower door **30** may be considered to be fully opened.

(226) When the lower door **30** is fully opened, the lower link assembly **80** may be fully unfolded. The first lower sub link **83** and the second lower sub link **84** may be supported from the lower side by the lower connection link **82** to prevent the first lower sub link **83** and the second lower sub link **84** from drooping. The user may close the lower door **30**, and while the lower door **30** is closed, the

lower link assembly **80** may operate in a reverse order of the above-described process.

(227) In addition to the foregoing embodiment, the multi-joint link hinge according to various embodiments may be exemplified. In another embodiment, a combination of the upper link module or assembly and the lower link module or assembly of the multi-joint link hinge is different, but other detailed configurations may be the same or similar. Thus, the same or similar components are denoted using the same reference numerals and detailed descriptions thereof may be omitted to prevent duplication of description.

(228) Referring to FIG. **36**, a multi-joint link hinge **60'** according to another embodiment may include a hinge bracket **61** fixed and mounted to a cabinet **10**, and an upper link assembly **70** and a lower link assembly **70'** arranged vertically inside the hinge bracket **61**. A structure and shape of the hinge bracket **61** and the upper link assembly **70** may be the same as or similar to those of the above-described embodiment.

(229) The hinge bracket **61** may have the same structure as the hinge bracket **61** according to the foregoing embodiment. A vertical width of the hinge bracket **61** may be predetermined or prescribed to receive both the upper link assembly **70** and the lower link assembly **70'**. Each of the upper link assembly **70** and the lower link assembly **70'** may have a thickness at which each of the upper link assembly **70** and the lower link assembly **70'** may be rotatable within an accommodation or receiving space inside the hinge bracket **61**. Each of the upper link assembly **70** and the lower link assembly **70'** may have a thickness less than that in the foregoing embodiment. The hinge bracket **61** may have a larger vertical width within a range in which the hinge bracket **61** is capable of being attached to a barrier **11**, and in this case, the lower link assembly **70'**, having the same structure and same or similar thickness as the upper link assembly **70** in this embodiment, may be mounted.

(230) The upper link assembly **70** may include an upper main link **71**, an upper connection link **72**, a first upper sub link **73**, a second upper sub link **74**, and an upper door bracket **75**, which are the same as those according to the previous embodiment.

(231) The lower link assembly **70'** may have the same structure as the upper link assembly **70**. The lower link assembly **70'** may also include a lower main link **71'** having a bent structure like the upper main link **71**, a first lower sub link **73'** provided inside the lower main link **71'**, a second lower sub link **74'**, and a lower connection link. These configurations may differ only in their names, and may have the same shape as the upper main link **71**, the first upper sub link **73**, the second upper sub link **74**, and the upper connecting link **72**, respectively.

(232) Since the lower link assembly **70'** may be connected to a top surface of the lower door **30**, a lower shaft of the lower door bracket **75'** may be provided to face a lower side. The lower link assembly **70'** may have the same structure and shape as the upper link assembly **70**, but a thickness of the lower main link **71'** may be slightly thinner than the thickness of the upper main link **71**.

(233) In another embodiment, a combination of the upper link module and the lower link module constituting the multi-joint link hinge may be different, but other detailed configurations may be the same or similar. Thus, the same components are denoted using the same reference numerals or detailed descriptions thereof will be omitted to prevent duplication of description.

(234) Referring to FIG. **37**, a multi-joint link hinge **60'** according to another embodiment may include a hinge bracket **61** fixed and mounted to a cabinet **10** and an upper link assembly **80** and a lower link assembly **80'** vertically arranged inside the hinge bracket **61**. A structure and shape of the hinge bracket **61** and the lower link assembly **80** may be the same as or similar to those of the above-described embodiment.

(235) The hinge bracket **61** may have an accommodation or receiving space in which the upper link assembly **80'** and the lower link assembly **80** are provided. Each of the upper link assembly **80'** and the lower link assembly **80** may have a plate shape, and a vertical thickness of the hinge bracket **61** may be thinner than that of the previous embodiments.

(236) When the thickness of the hinge bracket **61** is thinner, a multi-joint link hinge **60''** may be

more compact. A thickness of the barrier **11** may be thinner, and an space between an upper door **20** and a lower door **30** may be narrower to improve an outer appearance.

(237) The lower link assembly **80** may include a lower main link **81**, a lower connection link **82**, a first lower sub link **83**, a second lower sub link **84**, and a lower door bracket **85**, which are the same as or similar to those according to the previous embodiment of the lower link assembly **80**.

(238) The upper link assembly **80'** may have the same structure as the lower link assembly **80**. The upper link assembly **80'** may also include an upper main link **81'** having a plate-shaped structure like the lower link assembly **80**, and a first upper sub link **83'** coupled to the upper main link **81'**, a second upper sub link **84'**, and an upper connection link **82**. These configurations may differ only in their names and may have the same or similar shape as the lower main link **81**, the first lower sub link **83**, the second lower sub link **84**, and the lower connecting link **82**, respectively. Since the upper link assembly **80'** may to be connected to a bottom surface of the upper door **20**, an upper shaft of the upper door bracket **85'** may be provided to face an upper side.

(239) In addition to the foregoing embodiments, a refrigerator according to various embodiments may be exemplified. Such embodiment is different only in structure of the cabinet and the door, while configurations of the multi-joint link hinge may be the same as or similar to the previous embodiments. The same components are denoted using the same reference numerals and/or detailed descriptions thereof will be omitted to prevent duplication of description.

(240) Referring to FIGS. **38-39**, a refrigerator **1'** according to another embodiment may include a cabinet **10'** in which a storage space is defined and doors **20'** and **30'** that open and close the storage space. The storage space within the cabinet **10'** may be vertically partitioned by a barrier **11'** to provide a refrigerating compartment **12'** at an upper side and a freezing compartment **13'** at a lower side. The doors **20'** and **30'** may include a refrigerating compartment door **20'** and a freezing compartment door **30'**, which open and close the refrigerating compartment **12'** and the freezing compartment **13'**, respectively. The refrigerating compartment door **20'** and the freezing compartment door **30'** may be rotatably mounted to the cabinet **10'**.

(241) An upper hinge **40**, a lower hinge **50**, and a multi-joint link hinge **60** may be provided on a front surface of the cabinet **10'**. A structure of each of the upper hinge **40**, the lower hinge **50**, and the multi-joint link hinge **60** may be the same as or similar to those of the previously described embodiments, and thus detailed descriptions thereof will be omitted.

(242) The refrigerator **1'** may be provided to be embedded in furniture, fixtures, or a wall, or a plurality of refrigerators **1'** may be provided side by side or provided side by side with other home appliances. Since a periphery of the door **20'** and **30'** are provided very close to the furniture or the wall, other refrigerators, or home appliances, an interval therebetween may be substantially narrowed.

(243) The refrigerating compartment door **20'** and the freezing compartment door **30'** may rotate so as not to interfere with the adjacent furniture or walls, other refrigerators, or the home appliances. The upper hinge **40**, the lower hinge **50**, and the multi-joint link hinge **60** may be configured as a combination of a plurality of links, and while the upper hinge **40**, the lower hinge **50**, and the multi-joint link hinge **60** operate, the doors **20'** and **30'** may be configured to rotate along a set trajectory that does not interfere with the furniture or wall, other refrigerators, or home appliances.

(244) The multi-joint link hinge **60** may be mounted on the barrier **11'** to independently support the refrigerating compartment door **20'** and the freezing compartment door **30'**, and the refrigerating compartment door **20'** and the freezing compartment door **30'** may rotate by the multi-joint link hinge **60**. The multi-joint link hinge **60** may be mounted on the barrier **11'** having a narrow vertical width, and in particular, may have a compact structure and thus have a mounting structure in which a distance between a lower end of the refrigerating compartment door **20'** and an upper end of the freezing compartment door **30'** is minimized.

(245) Even when weights of the doors **20'** and **30'**, which respectively shield the refrigerating compartment **12'** and the freezing compartment **13'**, are heavy, drooping may not occur, and an

interference between upper link modules **70** and **70'** and lower link modules **80** and **80'**, which may be provided in the multi-joint link hinges **60**, **60'**, and/or **60''** may be prevented, facilitating a stable and smooth opening/closing operation.

(246) The structure of the multi-joint link hinge **60** connecting the cabinet **10** of the refrigerator **1** to the doors **20** and **30** has been described as an example, but are not limited thereto. For example, the structure of the multi-joint link hinge may be applied to other home appliances in addition to the refrigerator. When the doors are arranged vertically, the multi-joint link hinge **60** may be provided between an upper door and a lower door to be connected to the upper door and the lower door, respectively.

(247) Embodiments disclosed herein may provide a multi-joint link hinge. The multi-joint link hinge may have a structure in which an upper link module or assembly rotatably connects an upper door to one hinge bracket and a lower link module or assembly rotatably connects a lower door to one hinge bracket. The upper and lower link modules may be coupled to each other. The upper link module and the lower link module may be configured by coupling the plurality of links to allow the user to more easily open and close the upper and lower doors because a rotation of the doors do not interfere with the furniture, the walls, or the home appliances.

(248) Due to the link structure of the center hinge, when the upper door and the lower door are closed, a distance between the refrigerator and the furniture, the walls, or the home appliances may be reduced or minimized.

(249) The main link module may have relatively thicker thickness than the lower link module, and the upper link module and/or the lower link module supporting the upper door, which may carry a relatively higher load in the limited space of the center hinge, may have a thicker thickness to prevent the upper link module from being deformed or drooping even when used for a long time.

(250) The upper main link of the upper link module may be molded by bending the metal plate-like material to ensure or reinforce strength. While increasing in thickness of the upper main link, the upper connection link, the first upper sub link, and the second upper sub link may be received within the upper main link so that a strength in limited height specification may be satisfied, and thickness may be reduced or minimized.

(251) The upper link module may include the upper connection link supporting the first upper sub link and the second upper sub link, and the lower link module may include the connection link supporting the first lower sub link and the second lower sub link. An interference due to the drooping of the upper link and the lower link may be prevented even in the rotation structure of the upper door and the lower door, each of which has a long operation distance and the large load, by having the plurality of link structures according to the embodiment. The opening and closing operations of the upper door and the lower door may be secured without interfering with each other.

(252) The upper link module and the lower link module may be provided inside the hinge bracket in the folded state, and each of the links may have the tilted structure corresponding to each other so that the links have the hexahedral shape in the fully folded state. When the upper link module and the lower link module are folded, the size of each of the upper link module and the lower link module may be maintained in compact state, and due to this structure, the mounting space may be minimized or reduced, and the link modules may be prevented from being exposed to the outside.

(253) The upper link module and the lower link module may be provided in the region inside a vertical height of the hinge bracket, and a thickness of each of the link modules may be reduced or minimized by the bent structure of the upper main link and the lower main link structure. A center hinge may be implemented to be slimmer, and an interval between the upper door and the lower door on which the center hinge is mounted may be minimized to improve the outer appearance. Due to the slim center hinge structure, the exposure of the center hinge between the upper door and the lower door may be minimized to further improve the outer appearance.

(254) Since a vertical width or height of the center hinge may be reduced or minimized, the center

hinge may be easily mounted on the front surface of the barrier having the limited thickness. Since the center hinge may be provided in the space between heating members such as hot gas or a heater provided in the barrier, when assembled, the interference with the heating member may be prevented to significantly improve the workability.

(255) Since the thickness of the center hinge is reduced or minimized, there may be no need to secure the additional thickness of the barrier, and thus, the storage capacity of each of the upper storage space and the lower storage space may be prevented from being decreased by the barrier.

(256) Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

(257) Embodiments disclosed herein may include a multi-joint link hinge which prevents an interference with adjacent objects and easily opens and close an upper door and a lower door, and a refrigerator including the multi-joint link hinge. The multi-joint link hinge may prevent an upper door from drooping. The multi-joint link hinge may be capable of securing opening and closing operability of an upper door and a lower door.

(258) The multi-joint link hinge may have compact structure. The multi-joint link hinge may have a slim thickness. The multi-joint link hinge may have improved installation workability and convenience.

(259) Embodiments disclosed herein may be implemented as a multi-joint link hinge including a hinge bracket provided with a base portion, in which a plurality of screw coupling holes are defined, and a plurality of side portions, which protrude from the base portion and are spaced apart from each other in a vertical direction. A plurality of link modules may be vertically arranged between the side portions and axially coupled to the hinge bracket by a rotation shaft. A door bracket may be axially coupled to each of the plurality of link modules.

(260) The hinge bracket may be made of a metal material, and the side portions may be bent forward from upper and lower ends of the base portion, respectively. The multi-joint link hinge may further include a shielding portion that is bent forward from each of the ends of the base portion and connected to each of the side portions to shield a space between the side portions.

(261) The rotation shaft may be coupled to pass through the side portions and sequentially pass through the plurality of link modules. A spacer penetrated by the rotation shaft and configured to support the plurality of link modules may be provided between side portions so that a set or predetermined distance or interval between the plurality of link modules is maintained. A spacer insertion groove may be defined in a circumferential surface of the spacer, and each of the link modules may be penetrated by the rotation shaft in a state of being inserted onto the spacer insertion groove.

(262) Each of the plurality of link modules may be configured through coupling of a plurality of links. The rotation shaft may include a main rotation shaft and a sub rotation shaft. The main rotation shaft may be configured to pass through the side portions and one link of the plurality of links constituting the plurality of link modules. The sub rotation shaft may be configured to pass through the side portions, the spacer, and the other link of the plurality of links constituting the plurality of link modules at one side spaced apart from the main rotation shaft.

(263) The plurality of link modules may be configured as a combination of a plurality of links to connect the hinge bracket to the door bracket. The plurality of link modules may include an upper link module provided in a space between the side portions to support an upper door at a lower side, and a lower link module provided under the upper link module to support a lower door at an upper

side.

(264) The upper link module may have a thickness greater than that of the lower link module. Each of the upper link module and the lower link module may be configured through coupling of the links having the same length and number.

(265) The upper link module may include an upper main link that is axially coupled to the hinge bracket, a first upper sub link that is axially coupled to each of one side of the upper main link and one side of the door bracket, a second upper sub link that is axially coupled to each of the other side of the upper main link and the other side of the door bracket, and an upper connection link that is axially coupled to the hinge bracket, one side of the upper main link, and another side of the second upper sub link.

(266) The upper main link may be made of a plate-shaped metal material and may be bent several times to define a space in which the first upper sub link, the second upper sub link, and the connection link are accommodated or received. The upper main link may include a top surface and a bottom surface spaced apart from each other and a connection surface configured to connect one end of the top surface to one end of the bottom surface. The first upper sub link, the second upper sub link, and the upper connection link may be provided in a space between the top and bottom surfaces and the connection surface. The first upper sub link, the second upper sub link, and the connection link may be provided in the space in a state of being folded. The upper connection link may be provided under the first upper sub link and the second upper sub link and extend to support the first upper sub link and the second upper sub link at a lower side.

(267) The lower link module may include a lower main link that is axially coupled to the hinge bracket, a first lower sub link that is axially coupled to each of one side of the lower main link and one side of the door bracket, a second lower sub link that is axially coupled to each of the other side of the lower main link and the other side of the door bracket, and a lower connection link that is axially coupled to the hinge bracket, one side of the lower main link, and another side of the second lower sub link. The lower main link may have a plate shape having a thickness less than that of the upper main link.

(268) Embodiments disclosed herein may be implemented as a refrigerator including a cabinet in which an upper storage space and a lower storage space are defined, an upper door configured to open and close the upper storage space through rotation thereof, an upper hinge configured to connect an upper end of the upper door to the cabinet so that the upper door is rotatable, a lower door configured to open and close the lower storage space through rotation thereof, a lower hinge configured to connect a lower end of the lower door to the cabinet so that lower door is rotatable, and a multi-joint link hinge provided between the upper hinge and the lower hinge. The multi-joint link hinge may include a hinge bracket provided with a base portion, in which a plurality of screw coupling holes are defined to be coupled to a cabinet, and a plurality of side portions which protrude from the base portion and are spaced apart from each other in a vertical direction. The multi-joint link hinge may further include a plurality of link modules vertically provided between the side portions and axially coupled to the hinge bracket, and a door bracket which is axially coupled to each of the plurality of link modules and coupled to a door configured to open and close the cabinet.

(269) The plurality of link modules may be configured as a combination of a plurality of links to connect the hinge bracket to the door bracket, and the plurality of link modules may include an upper link module provided between the side portions to support the upper door at a lower side, and a lower link module provided under the upper link module to support the lower door at an upper side. The upper link module may have a thickness greater than that of the lower link module.

(270) The cabinet may include a barrier configured to partition the upper storage space and the lower storage space from each other, and the hinge bracket may be coupled to a front surface of the barrier. A heating member passing through the barrier may be provided on a circumference or outer side of each of the upper storage space and the lower storage space, and the hinge bracket may be

mounted between the heating members that are provided vertically. A distance between a front lower end of the upper door and a front upper end of the lower door may be less than a thickness of the multi-joint link hinge.

(271) Embodiments disclosed herein may be implemented as a refrigerator including a multi-joint link hinge. The multi-joint link hinge may include a hinge bracket provided between an upper door and a lower door, an upper link module having a plurality of links and configured to rotatably connect the hinge bracket to a lower end of the upper door, and a lower link module having a plurality of links and configured to rotatably connect the hinge bracket to an upper end of the lower door.

(272) Embodiments disclosed herein may be implemented as a refrigerator including a multi-joint link hinge. The multi-joint link hinge may include a hinge bracket provided between an upper door and a lower door, an upper link module having a plurality of links and configured to rotatably connect the hinge bracket to a lower end of the upper door, a lower link module having a plurality of links and configured to rotatably connect the hinge bracket to an upper end of the lower door. The upper link module and the lower link module may be vertically provided, and the upper link module may be a thicker thickness.

(273) Embodiments disclosed herein may be implemented as a refrigerator including a multi-joint link hinge. The multi-joint link hinge may include a hinge bracket mounted on a cabinet and an upper link module and a lower link module, which may be mounted on the hinge bracket and having a plurality of links. The upper link module and the lower link module may include a main link connected to the hinge bracket, a door bracket connected to an upper door or a lower door, and a first sub link and a second sub link, which may be configured to connect the main link to the door bracket. A connection link may be further coupled to the hinge bracket, and the connection link may be configured to support the first sub link and the second sub link at a lower side.

(274) Embodiments disclosed herein may be implemented as a refrigerator including a multi-joint link hinge. The multi-joint link hinge may include a hinge bracket coupled to a cabinet, an upper link module having a plurality of links and configured to rotatably connect the hinge bracket to a lower end of the upper door, and a lower link module having a plurality of links and configured to rotatably connect the hinge bracket to an upper end of the lower door. The upper link module and the lower link module may be received in the hinge bracket in a state of being completely folded.

(275) Embodiments disclosed herein may be implemented as a refrigerator including a multi-joint link hinge. The multi-joint link hinge may include a hinge bracket coupled to a cabinet, an upper link module having a plurality of links and configured to rotatably connect the hinge bracket to a lower end of the upper door, and a lower link module having a plurality of links and configured to rotatably connect the hinge bracket to an upper end of the lower door. The upper link module and the lower link module may be vertically provided inside the hinge bracket.

(276) Embodiments disclosed herein may be implemented as a refrigerator including a multi-joint link hinge. The multi-joint link hinge may include a hinge bracket coupled to a barrier, an upper link module having a plurality of links and configured to rotatably connect the hinge bracket to a lower end of the upper door, and a lower link module having a plurality of links and configured to rotatably connect the hinge bracket to an upper end of the lower door. The hinge bracket may have a size less than a distance between a pair of heating members provided in the barrier and is provided between the pair of heating members.

(277) A direction facing a front surface of the door illustrated in FIG. 1 may be defined as a front direction, a direction facing the inside of the refrigerator with respect to the front surface of the door may be defined as a rear direction, a direction facing a bottom surface on which the refrigerator may be installed may be defined as a downward direction, and a direction that may be away from the bottom surface may be defined as an upward direction.

(278) Embodiments disclosed herein may be implemented as a multi-joint link hinge comprising a first bracket, a first linkage, a second linkage, at least one rotation shaft, and a second bracket. The

first bracket may include a base plate having a plurality of screw holes and first and second side walls extending from the base plate and spaced apart from each other in a first direction. The first and second linkages may be arranged in the first direction and configured to be received between the first and second side walls. The at least one rotation shaft may rotatably couple the first and second linkages to the first bracket. The second bracket may be rotatably coupled to the first linkage.

(279) The first bracket may be made of a metal material. The first and second side walls may extend in a second direction from first and second sides of the base plate, respectively, the first and second sides being spaced apart in the first direction. A cover may extend from the base plate and be connected to each of the first and second side walls to at least partially cover a space between the first and second side walls.

(280) The rotation shaft may pass through the first and second side walls and the first and second linkages. The rotation shaft may be inserted through a space. The spacer may be configured maintain a prescribed distance between the first and second linkages.

(281) An outer surface of the spacer may include a groove. The first linkage may include a first connection link configured to be inserted into the spacer groove.

(282) The at least one rotation shaft may include a main rotation shaft and a sub rotation shaft. The main rotation shaft may be configured to pass through the side walls, a first main link of the first linkage, and a second main link of the second linkage. The sub rotation shaft may be configured to pass through the side walls, the spacer, a first connection link of the first linkage, and a second connection link of the second linkage. The main and sub rotation shafts may be spaced apart from each other.

(283) The first direction may be a vertical direction and the first linkage may be provided above the second linkage. The first linkage may be configured to couple to a first door via the second bracket, and the second linkage being configured to couple to a second door provided below the first door via a third bracket.

(284) The first linkage may have a thickness greater than that of the second linkage. The first linkage may have a plurality of first links, the second linkage may have a plurality of second links, a number of first links may be equal to a number of second links, and lengths of the respective first links are equal to lengths of the respective second links.

(285) The first direction may be a vertical direction. The first linkage may be provided above the second linkage. The first linkage may include an upper main link that may be axially coupled to the first bracket, a first upper sub link that may be axially coupled to a first side of the upper main link and axially coupled to a first side of the second bracket, a second upper sub link that may be axially coupled to a second side of the upper main link and axially coupled to a second side of the second bracket, and an upper connection link that may be axially coupled to the first bracket, the first side of the upper main link, and a side of the second upper sub link.

(286) The upper main link may be made of a plate-shaped metal material and may be bent several times to define a space in which the first upper sub link, the second upper sub link, and the connection link are received and arranged in a vertical direction.

(287) The upper main link may include a top surface, a bottom surface provided below the top surface, and a connection surface connecting the top surface to the bottom surface. The first upper sub link, the second upper sub link, and the upper connection link may be provided between the top and bottom surfaces. The first upper sub link, the second upper sub link, and the connection link may be configured to be provided in a space defined by the top, bottom, and connection surfaces in a folded state.

(288) The upper connection link may be provided under the first upper sub link and the second upper sub link. The upper connection link may be configured to extend to support lower sides of the first upper sub link and the second upper sub link.

(289) The second linkage may include a lower main link that may be axially coupled to the first

bracket, a first lower sub link that may be axially coupled to a first side of the lower main link and a first side of a third bracket, a second lower sub link that may be axially coupled to a second side of the lower main link and a second side of the third bracket, and a lower connection link that may be axially coupled to the first bracket, the first side of the lower main link, and a side of the second lower sub link. The lower main link may have a plate shape having a thickness less than that of the upper main link.

(290) Embodiments disclosed herein may be implemented as a refrigerator comprising a cabinet having an upper storage space and a lower storage space, an upper door configured to open or close the upper storage space, an upper hinge configured to rotatably connect a top of the upper door to the cabinet, a lower door configured to open or close the lower storage space, a lower hinge configured to rotatably connect a bottom of the lower door to the cabinet, and a multi-joint link hinge provided between the upper hinge and the lower hinge. The multi-joint link hinge may comprise a hinge bracket provided with a base plate and first and second side walls extending from the base plate to be spaced apart from each other in a vertical direction, the base plate being configured to be coupled to the cabinet, an upper linkage rotatably coupled to the upper door, and a lower linkage provided below the upper linkage and rotatably coupled to the lower door, the upper and lower linkages configured to be provided between the first and second side walls and axially coupled to the hinge bracket.

(291) The upper linkage may be configured to support the upper door at a lower side. The lower linkage may be configured to support the lower door at an upper side. The upper linkage may have a thickness greater than that of the lower linkage.

(292) The cabinet may include a barrier configured to partition the upper storage space from the lower storage space. The hinge bracket may be coupled to the barrier.

(293) A first heating member may pass through the barrier to surround the upper storage space. A second heating member may pass through the barrier to surround the lower storage space. The second heating member may be vertically spaced apart from the first heating member. The hinge bracket may be mounted between the first and second heating members.

(294) A distance between a front lower end of the upper door and a front upper end of the lower door may be less than a thickness of the multi-joint link hinge.

(295) Embodiments disclosed herein may be implemented as a home appliance having a multi-joint link hinge. The home appliance may include a cabinet having an upper storage space and a lower storage space, an upper door configured to open or close the upper storage space, an upper hinge configured to rotatably connect a top of the upper door to the cabinet, a lower door configured to open or close the lower storage space, and a lower hinge configured to rotatably connect a bottom of the lower door to the cabinet. The multi-joint link hinge may be provided between the upper hinge and the lower hinge.

(296) It will be understood that when an element or layer is referred to as being “on” another element or layer, the element or layer can be directly on another element or layer or intervening elements or layers. In contrast, when an element is referred to as being “directly on” another element or layer, there are no intervening elements or layers present. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(297) It will be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section could be termed a second element, component, region, layer or section without departing from the teachings of the present invention.

(298) Spatially relative terms, such as “lower”, “upper” and the like, may be used herein for ease of description to describe the relationship of one element or feature to another element(s) or feature(s)

as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “lower” relative to other elements or features would then be oriented “upper” relative to the other elements or features. Thus, the exemplary term “lower” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(299) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(300) Embodiments of the disclosure are described herein with reference to cross-section illustrations that are schematic illustrations of idealized embodiments (and intermediate structures) of the disclosure. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments of the disclosure should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

(301) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(302) Any reference in this specification to “one embodiment,” “an embodiment,” “example embodiment,” etc., means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of such phrases in various places in the specification are not necessarily all referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with any embodiment, it is submitted that it is within the purview of one skilled in the art to effect such feature, structure, or characteristic in connection with other ones of the embodiments.

(303) Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. A refrigerator comprising: a cabinet having an upper storage space and a lower storage space; an upper door configured to open or close the upper storage space; an upper hinge configured to rotatably connect a top of the upper door to the cabinet; a lower door configured to open or close the lower storage space; a lower hinge configured to rotatably connect a bottom of the lower door to the cabinet; and a center hinge provided between the upper hinge and the lower hinge, wherein

the center hinge comprises: a hinge bracket mounted at the cabinet; an upper link module rotatably coupled to the hinge bracket and configured to rotate the upper door relative to the cabinet; and a lower link module disposed below the upper link module and rotatably coupled to the hinge bracket and configured to rotate the lower door relative to the cabinet, wherein the upper link module comprises: a plurality of upper links rotatably coupled to the hinge bracket; and an upper door bracket mounted to a lower surface of the upper door, and coupled to at least one of the plurality of upper links, wherein the lower link module comprises: a plurality of lower links disposed below the upper link module and rotatably coupled to the hinge bracket; and a lower door bracket mounted to an upper surface of the lower door, and coupled to at least one of the plurality of lower links, wherein the plurality of lower links comprise: a lower main link rotatably coupled to the hinge bracket, and the lower main link faces a bottom surface of the upper link module; a first lower sub link rotatably coupled to one side of the lower main link and one side of the lower door bracket; a second lower sub link rotatably coupled to another side of the lower main link and another side of the lower door bracket; and a lower connection link rotatably coupled to the lower main link and the second lower sub link, and wherein a support is disposed on an upper surface of the lower main link, the support is disposed between shafts that separately pass through at least two of the lower links, and the support selectively contacts with a lower surface of the upper link module depending on a load of the upper door.

2. The refrigerator according to claim 1, wherein the support is in contact with the upper link module when the upper link module downward droops during rotation of the upper link module, and the support is made of material having abrasion resistance and lubricity.

3. The refrigerator according to claim 2, wherein the support includes a plurality of supporters spaced apart from each other on the upper surface of the lower link module.

4. The refrigerator according to claim 2, wherein the support is made of engineering plastic or polyoxymethylene (POM) material.

5. The refrigerator according to claim 1, wherein the lower main link, the first lower sub link, the second lower sub link, and the lower connection link are each separately made based on a plate-shaped metal material.

6. The refrigerator according to claim 1, wherein the upper link module comprises: an upper main link rotatably coupled to the hinge bracket; a first upper sub link rotatably coupled to a first side of the upper main link, and the first upper sub link is rotatably coupled to a first side of the upper door bracket; a second upper sub link rotatably coupled to a second side of the upper main link, and the second upper sub link is rotatably coupled to a second side of the upper door bracket; and an upper connection link rotatably coupled to the hinge bracket, the first side of the upper main link, and the second upper sub link.

7. The refrigerator according to claim 6, wherein the upper main link is made based on a plate-shaped metal material that is bent a plurality of times so to define a space configured to receive the first upper sub link, the second upper sub link, and the upper connection link which are arranged in a vertical direction.

8. The refrigerator according to claim 7, wherein a thickness of the upper link module is greater than a thickness of the lower link module.

9. The refrigerator according to claim 1, wherein a distance between a front lower end of the upper door and a front upper end of the lower door is less than a thickness of the center hinge, and at least a portion of the center hinge is covered by the upper door or the lower door.

10. The refrigerator according to claim 9, wherein the upper surface of the lower door includes a recess to accommodate a lower part of the center hinge.

11. The refrigerator according to claim 10, wherein the recess is configured to accommodate at least a portion of the lower link module.

12. The refrigerator according to claim 11, wherein a lower end of the upper link module is disposed higher than the recess.

13. The refrigerator according to claim 1, wherein the upper door includes a frame configured to define a bottom surface of the upper door, and the frame includes an upper link mounting recess to couple with the upper door bracket.

14. The refrigerator according to claim 1, wherein the lower door includes a frame configured to define a top surface of the lower door, and the frame includes a lower link mounting recess to couple with the lower door bracket.

15. The refrigerator according to claim 1, wherein the hinge bracket includes: a base mounted to a front surface of the cabinet; a first wall that extends from an upper edge of the base; and a second wall that extends from a lower edge of the base, and the second wall is spaced apart from the first wall, and wherein the upper link module and the lower link module are rotatably coupled to the first and second walls.

16. The refrigerator according to claim 15, wherein the hinge bracket includes a receiving space between the first wall and the second wall to accommodate the upper link module and the lower link module when the upper door and the lower door are closed.

17. The refrigerator according to claim 15, wherein the hinge bracket includes a shield disposed at an end of the base and is coupled to the first wall and the second wall to cover part of the upper link module and part of the lower link module.

18. The refrigerator according to claim 15, wherein a shaft penetrates the first wall, the upper link module, the lower link module, and the second wall so as to rotatably couple the upper link module and the lower link module to the hinge bracket.

19. A refrigerator comprising: a cabinet having an upper storage space and a lower storage space; an upper door configured to open or close the upper storage space; an upper hinge to couple a top of the upper door and the cabinet; a lower door configured to open or close the lower storage space; a lower hinge to couple a bottom of the lower door and the cabinet; and a center hinge provided between the upper hinge and the lower hinge, wherein the center hinge includes: a hinge bracket, an upper link module rotatably coupled to the hinge bracket and configured to rotate the upper door relative to the cabinet, and a lower link module rotatably coupled to the hinge bracket and configured to rotate the lower door relative to the cabinet; wherein the upper link module includes: an upper door bracket to mount to the upper door, and a plurality of upper links configured to provide rotation of the upper door, wherein the lower link module includes: a lower door bracket to mount to the lower door, and a plurality of lower links to provide rotation of the lower door, the plurality of lower links including a lower main link having a plate shape, a first lower sub link, a second lower sub link, and a lower connection link, the first lower sub link and the second lower sub link are rotatably connected to each other between the lower main link and the lower door bracket, and the lower connection link connecting the second lower sub link to the hinge bracket, wherein at least two lubricity supports are disposed on part of a surface of the lower main link having the plate shape and the at least two lubricity supports to face the upper link module when the upper door and the lower door are both closed, and the at least two lubricity supports are to physically contact the upper link module and thereby provide space between the surface of the lower main link and the upper link module during rotation of the upper link module.
